Supplementary information: A common BSA–serum equation for EASI estimation in a Chinese atopic dermatitis cohort

Supplementary guide

The Supplementary Methods define the analysis populations, endpoints and validation. Tables S1–S5 provide exact target rates, complete model metrics, safety-documentation coverage, reference-exclusion sensitivity and input/reference specifications. The 41 figures are grouped below; complete aggregate results, including negative and non-estimable findings, remain in the linked CSV collection. Missing estimates are not zero effects. **Cohort structure and treatment comparability**. Parallel patient populations, sequential exclusions, baseline balance, common support and actual recorded exposure.

Clinical outcomes and stage-specific targets. Complete specified visits, observed and adjusted trajectories, IGA composition, response targets, dose-reduction pathways and all treatment contrasts.

The full molecular panel and regional disease. All 23 markers, paired change windows, clinical associations, response-depth profiles and regional analyses.

Equation validation and other targets. Transport and severity checks, separately fitted PP-NRS/DLQI/IGA targets, and response classification. Matched numerical controls remain in source tables.

D denotes dupilumab; A100/A200 denote initial abrocitinib 100/200 mg. Treatment colours are fixed throughout. Molecular changes use $\log_2$ concentration units; a change of -1 means a halving. Model errors use EASI points. Correlation describes ordering, whereas absolute errors and concordance address numerical agreement.

1 Supplementary Methods

1.1 Study design and clinical records

This analysis used longitudinal records from a Chinese observational treat-to-target project. The supplied protocol specified clinician-directed systemic treatment rather than random allocation. We retained the initial treatment group for longitudinal comparisons and separately extracted subsequent dose, interval, stopping, switching and restart information. The analysed records came from CAMS and Huashan. The protocol's planned centre list and planned sample size were not substituted for the observed cohort.

The clinical dose-comparison population included patients aged at least 12 years with a known initial regimen. Records with unresolved simultaneous initial-dose assignments were excluded from dose-specific comparisons but retained in the source audit. Adult DLQI analyses used age at least 18. Repeated records were linked by the study patient identifier and nominal visit; a patient was never treated as a new independent person at each visit. The requested reporting visits were W0, W2, W4, W12 and W24 for PP-NRS; W0, W4, W12 and W24 for EASI, IGA, DLQI, BSA, SCORAD and eosinophils; and W0, W12 and W24 for IgE. Low-coverage intermediate visits were retained in the underlying data but were not inserted into these trajectories.

EASI was the recorded clinical reference outcome. Source discrepancies between regional sums and total EASI were audited rather than silently repaired. IGA grades were retained as recorded, including grade 5. Confirmation of the exact IGA version and PP-NRS recall window remains necessary. Nominal clinical/serum visit linkage does not by itself verify that the blood draw and examination occurred on the same calendar day; exact dates and a permissible matching window require source confirmation. Among the 152/89/75 model visits at W0/W12/W24, 8/2/0 had complete regional scores whose sum differed from total EASI by more than 0.15; the largest absolute gaps were 18.7/8.5/0.0 points, and 1/1/4 visits lacked a complete regional reference. Restricting evaluation to all four regions observed with total–sum difference at most 0.01 retained 140/86/71 visits and gave MAEs 3.17/1.32/1.11. This evaluation-only sensitivity uses unchanged saved predictions. A separate training-exclusion sensitivity removed the eight W0 and two W12 discrepant visits from fitting, retained missing-region visits, and repeated preprocessing and inner structure/penalty selection within the original patient folds. Evaluation patients and recorded outcomes were unchanged. It gave W0/W12/W24 MAEs of 3.19/1.15/1.06; matched BSA controls were refitted identically. Neither analysis adjudicates which clinical reference is correct (Supplementary Table S4). Details are in `model_reference_QC_audit.csv`.

1.2 Clinical targets and treatment comparisons

The week-2 target was PP-NRS ≤ 4. At week 4, the physician domain was EASI50 or an IGA decrease of at least two points, and the patient domain was PP-NRS ≤ 4 or adult DLQI ≤ 10. At week 12, the physician domain was EASI75 or IGA 0/1, and the patient domain was PP-NRS 0/1 or adult DLQI ≤ 4. At week 24, EASI ≤ 5 was an additional physician-domain criterion. The combined target required both domains. Because instrument identity and recall windows remain unresolved, these are exploratory targets based on the recorded scores. These are the investigators' operational definitions; they were not retrospectively relabelled as the newer IEC low-disease-activity or remission definitions.[1]

For an OR criterion, one observed positive component established success; a negative result required all components to be observed and negative. For an AND criterion, one observed negative component established failure; success required both positive components. Other cases remained unknown. Adult DLQI was an unknown component for ages 12–17, not a negative component and not removed from the OR rule. Thus adolescent itch success establishes the patient target; itch failure alone does not establish patient-target failure. The audit exhaustively verified the nine two-component truth states and reconciled every saved composite. Among 15 adolescents, the patient domain had 6/0/9 successes/failures/unknowns at W4, 4/0/11 at W12 and 4/0/11 at W24; the combined domain had 5/1/9, 4/5/6 and 4/3/8. An explicitly different PP-NRS-only adolescent rule would make five W12 and four W24 patient-domain unknowns determinate, including one W24 combined outcome. This is a sensitivity count, not a redefinition of the primary composite. Age-stratified counts are in `composite_age_evaluability_audit.csv`. EASI response required an observed positive baseline. Individual thresholds were also analysed separately. Proportions used endpoint-specific observed denominators and Wilson 95% confidence intervals. Overall three-group tests used conditional Monte Carlo Pearson tests with 49,999 draws, and pairwise comparisons used Fisher's exact test and risk-difference intervals. The three pairwise comparisons within an endpoint and visit formed a Holm family; exploratory FDR results were also retained. Overall tests were not included as an extra member of that pairwise family.

Baseline-adjusted binary target rates and risk differences used generalized overlap weights from a multinomial initial-treatment model with baseline EASI, age, sex, BMI and centre. We used 1,000 patient bootstrap resamples, stratified by initial treatment and refitting the propensity model, for percentile intervals. Two-sided P values used normal Wald tests based on bootstrap standard errors; three-group tests used a two-degree-of-freedom Wald

statistic. The three individual pairwise comparisons formed a Holm family. Weighted results supplement the observed numerator and denominator. The primary target analysis used bootstrap seed 20261017. A revision W2 sensitivity added baseline PP-NRS to the propensity model. One patient lacked baseline PP-NRS; both the original-covariate model and the added-itch model were fitted on the remaining 162 patients, with the original 163-patient analysis retained. The same standardized logistic model (ridge parameter $C = 1$), 1,000 stratified patient resamples and seed 20261018 were used for this sensitivity. The revision files also contain a separate full-cohort run with seed 20261018. Its point estimates use the same analysis, but its Monte Carlo intervals and Wald P values are not the frozen primary results; borderline changes were not selected for the main claim. The added-itch 100 mg–dupilumab contrast was 41.4 percentage points (95% CI, 13.0–66.8; three-contrast Holm $P = 0.0104$); the 100-minus-200 mg contrast was 6.7 points (−11.3 to 26.2). The largest endpoint-specific absolute standardized difference was 0.53 even after adding itch, so residual imbalance remained. Baseline weights were retained when calculating available-case balance and ESS; no claim of balanced follow-up selection is made. The table below reports both unweighted and matching baseline-overlap-weighted worst/best bounds, assigning every unknown outcome failure/success while retaining baseline weights. These bound the initial cohort under extreme missing-outcome assignments; the reported weighted available-case rate represents patients with that endpoint observed. Endpoint-specific overlap-weight ESS is reported separately. Bounds are sensitivity limits, not confidence intervals.

Unweighted observed outcomes and initial-cohort bounds

Target	Initial group	Success/evaluable	Initial N	Observed rate, %	Unweighted missing-outcome bounds, %
W2 itch	D	15/27	65	55.6	23.1–81.5
W2 itch	A100	34/37	56	91.9	60.7–94.6
W2 itch	A200	33/36	42	91.7	78.6–92.9
W24 combined	D	26/36	65	72.2	40.0–84.6
W24 combined	A100	22/24	56	91.7	39.3–96.4
W24 combined	A200	20/23	42	87.0	47.6–92.9

Matching baseline-overlap-weighted analysis

Target	Initial group	Weighted available-case rate, %	Available-case weight ESS	Weighted missing-outcome bounds, %
W2 itch	D	51.9	17.6	19.4–82.0
W2 itch	A100	93.6	25.1	76.0–94.8
W2 itch	A200	85.8	21.1	63.3–89.5
W24 combined	D	46.2	20.8	22.0–74.3
W24 combined	A100	85.4	14.7	26.2–95.5
W24 combined	A200	87.0	12.5	38.3–94.3

Bounds assign all unknown outcomes failure or success in the initial cohort. Weighted bounds retain the same baseline overlap weights in observed and unknown records. Weighted available-case rates use only endpoint-observed patients. ESS is the effective sample size of those weighted available cases, not a patient count or an incidence denominator. Bounds are sensitivity limits, not confidence intervals.

For generalized propensity scores $e_t(x)$, the weight is $h(x)/e_t(x)$, with $h(x) = 1/\sum_t[1/e_t(x)]$. The baseline weighting function defines a common treatment-overlap population; an available-outcome analysis can depart from that population through selective observation. Covariate balance, propensity distributions and analysis-specific effective sample sizes are reported; weighting does not establish elimination of confounding.[2]

Continuous treatment comparisons used marginal regression with baseline outcome, treatment-by-visit terms, age, sex, BMI and centre. Generalized overlap weights addressed measured baseline differences in initial treatment selection. Patient-cluster robust covariance accounted for repeated outcomes, and a singular-value decomposition check prevented reporting non-estimable treatment-by-visit cells. IgE and eosinophils used $\log(1 + x)$; back-transformed estimates are not arithmetic means. Unweighted adjustment, group balance, observation patterns and supplementary dupilumab-versus-pooled-abrocitinib contrasts were retained. Pooling used the observed dose mixture and did not erase the two initial abrocitinib groups. The comparisons estimate adjusted observational associations, not randomized fixed-dose effects. Nominal eligibility based on the registry filename date placed all initial patients beyond W24, but this administrative sensitivity is not an approved database lock or verified visit window. Reasons for absent assessments were unavailable.

Treatment reductions were assigned from recorded regimen intervals, with assessment/decision ordering marked uncertain at shared visit boundaries. A later reduction was not moved to an earlier visit. W12 response and subsequent regimen changes defined descriptive maintenance pathways. Analyses used observed W24 EASI75, explicit missing-outcome bounds and fixed-patient trajectories where all three visits were observed. This approach distinguishes treatment initiation from the selected maintenance pathways, consistent with the need to specify how intercurrent treatment changes enter the question being estimated.[3] It does not identify a counter-

factual outcome under a different regimen. At W0, two source records labelled initial 100 mg explicitly listed a 200 mg starting regimen, and one labelled initial 200 mg listed 100 mg. These are conflicts between the source group and regimen fields, not demonstrated post-baseline switches. The group labels were retained, and `exposure_W0_source_adjudication.csv` preserves their counts separately from unknown exposure. Clinical-record adjudication is still required.

1.3 Serum measurements and molecular analyses

The laboratory source reports identify serum samples analysed with Luminex and MSD platforms. Luminex used an R&D Systems LXSAHM-13 panel and Luminex-200; the MSD reports identify K15067M-1/K15067M-2 kits and an SQ120 instrument. Reports describe 50-μL serum and twofold dilution. The service-batch sample count was not used as the study sample size. Assay lots, run identifiers and source links remain in the restricted provenance records.

The analysis used the numeric integration export rather than the vendor's filtered display matrix. Positive export numerics recorded in pg/mL were $\log_2$ transformed once, unchanged; the analysis applied no additional twofold dilution correction. This reproduces the fitted scale rather than verifying how the laboratory corrected its export for dilution. Explicit censoring strings and nonpositive values were not converted to measured positive concentrations. The vendor's filtered output can replace values below range with zero and values above range with an upper bound; those conventions must not be interpreted as exact measured concentrations. A sensitivity analysis represented IL-5 values below twice the software calibration lower bound at that boundary and repeated the model workflow. The software boundary is not an independently validated clinical quantification limit. Cross-platform bridging, technical-repeat precision and lot effects require additional assay documentation.

All 23 measured markers were retained. Molecular changes were formed within each patient and visit pair before group summaries, avoiding subtraction of averages from different patient sets. We analysed W0–W12, W12–W24 and W0–W24 separately. Mean $\log_2$ changes were accompanied by t-based confidence intervals when at least five pairs were available. Smaller sets were descriptive. Two-sided paired mean-change sign-flip tests retained zero changes and ties. They enumerated all 2^n signs when $n \leq 15$; otherwise they used 19,999 random signs (seed 20261002), counting exceedances at the observed absolute mean with a plus-one numerator and denominator. All-zero changes gave $P = 1$. The sign-flip null assumes independent patients and sign-exchangeable paired changes (symmetry about zero); it is not a randomized-treatment test. Benjamini–Hochberg adjustment covered the full 23-marker, three-treatment, three-window family: 207 planned cells, of which 193 had estimable P values. Non-estimable cells remained missing and were not counted as tests. Between-treatment change models adjusted for starting concentration, baseline EASI, age, sex, BMI and centre, with HC3 covariance. Centre-specific and assay-batch sensitivity analyses were retained. Each between-treatment molecular correction family included all marker, window and pairwise contrasts within one centre scope and analysis variant; scopes and variants were not pooled. Molecular–clinical change correlations formed their own panel-wide family across markers, clinical outcomes, treatments and windows, and response-depth contrasts formed a separate family. The global molecular treatment-by-visit family had 23 planned marker tests, with 21 estimable. A source-table inventory records the defined families and finite-test counts. Spearman correlations related molecular changes to clinical changes within treatment groups; baseline regional correlations were assessed separately. Sample sizes, intervals and multiplicity families accompany each source table.

Regional analyses used source-consistent regional totals. Regional EASI contributions were divided by their theoretical maxima (7.2, 14.4, 21.6 and 28.8 for head/neck, upper limbs, trunk and lower limbs). Percentage improvement required a positive regional starting value; zero-start exclusions were counted. Because very low W12 values can produce extreme W12–W24 percentage changes, the corresponding figure instead shows change as a percentage of the theoretical regional maximum. Both definitions are retained in the tables. Regional treatment contrasts used unadjusted Welch comparisons and HC3 models adjusted for starting regional severity, baseline total EASI, age, sex, BMI and centre, with FDR over the 48 region-by-window-by-contrast tests in each analysis/outcome family. Whole-body PP-NRS change was not labelled as regional itch.

1.4 Common EASI estimator and validation

Eligibility required age at least 12, observed EASI and BSA, and observed positive CCL17, CCL26 and IL-5 concentrations. Ambiguous initial-dose labels did not exclude otherwise eligible model records, because treatment was not a required input. The three-marker panel arose from earlier exploratory work and was fixed for this revision. No clinical outcome was imputed. Model development used W0 and W12; W24 did not enter fitting or selection. Because earlier project analyses had already examined W24, its results are described as within-cohort temporal application rather than untouched external validation.

All observations from one patient remained in the same outer fold. Five-fold validation was repeated twice using the established patient partitions. Each training fold contained development visits only. Three-fold patient-level inner validation selected the ridge penalty from 0.1, 1, 10 and 100, minimizing mean per-patient absolute error. Each patient received equal total fitting weight. Feature centring and scaling used training data only. Three common structures were considered: an additive BSA-plus-three-marker model; the same model with quadratic BSA; and BSA multiplied by a linear molecular component without a free intercept. Structure and penalty selection were nested together for the primary performance estimate. No treatment, phase, time, previous score, age or BMI was required at deployment.

After full-development inner selection, the fitted common equation was

$$\widehat{\mathrm{EASI}} = \mathrm{clip}_{[0,72]}\{\mathrm{BSA}\,[0.512721579 + 0.009847080\log_2(\mathrm{CCL17}) + 0.009646929\log_2(\mathrm{CCL26}) - 0.008270558\log_2(\text{IL-5})]\}.$$

BSA is entered as a percentage from 0 to 100, and concentrations must use the development export's recorded pg/mL numeric scale, with $\log_2$ applied once. The implementation rejects missing, nonfinite or nonpositive molecular inputs and invalid BSA values. The area constraint returns zero when BSA is zero. It is a statistical constraint reflecting absent observed lesion extent, not proof of absent molecular inflammation. The released parameter file retains full precision. The main-text display multiplies the molecular coefficients by 100 and uses BSA/100, allowing two decimal places without dropping the smaller molecular terms. An explicit rounding audit gave nested-selection W0/W12/W24 MAEs of 3.194/1.298/1.193, compared with 3.194/1.297/1.192 at full precision. All primary results use full precision; the rounded equation is a reading aid. For the final fitted equation, maximum absolute arithmetic differences were 0.075550, 0.024807 and 0.018474 EASI points at W0/W12/W24; these are rounding differences, not clinical prediction errors. Response classifications at EASI50/75/90 were unchanged in the audit, but one of 150 week-24 fold-level evaluations crossed the absolute EASI 7 threshold, so threshold-based use should retain full precision. This single final fit has not received independent validation; primary metrics evaluate the nested development procedure that selected it.

Comparators used the same patients and folds: BSA alone, the three markers alone, the additive common model, and the earlier stage-interaction model. We evaluated MAE, RMSE, signed error, Spearman correlation, Lin's concordance correlation, R^2, calibration intercept and slope, severity-band agreement and proportions within one and two EASI points. Metrics were calculated within each repetition before averaging, not from averaged predictions. Patient bootstrap intervals used 2,000 resamples of held-out errors. The same sampled patient indices were used in every repetition and comparator; metrics were scored within each repetition before averaging. These intervals describe uncertainty conditional on the saved cross-validation predictions: neither nested selection nor fitting was repeated, and earlier adaptive marker exploration is not covered. Calibration intervals resampled patients in the regression of observed on predicted EASI. The primary nested-procedure minus area-only MAE differences were 0.12 (95% CI, 0.04–0.20) at W12 and 0.17 (0.11–0.24) at W24. These paired estimates differ from the earlier fixed-structure comparison and do not establish molecular added accuracy. All comparator and fixed-structure results remain in the source tables. Descriptive agreement limits were mean error plus or minus 1.96 error standard deviations.[4]

Severity-band agreement used four categories: EASI ≤ 7, > 7 to ≤ 21, > 21 to ≤ 50 and > 50. Error subgroup analyses instead combined the two highest categories, using three strata: EASI ≤ 7, > 7 to ≤ 21 and > 21. Binary sensitivity and specificity used EASI > 7 and are not interchangeable with every original p-EASI classification definition. Leave-centre-out and leave-initial-treatment-group-out analyses removed all records from held-out patients. Changes were paired within patient, repetition and fold before averaging. Each of W0–W12, W0–W24 and W12–W24 was evaluated separately; the windows share patients. EASI50/75/90 classification required positive observed and estimated starting scores, with exclusions counted. Observed response used observed scores at both visits, and estimated response used estimated scores at both visits. Scatter plots and any integer confusion matrices use the first outer repetition; the primary response forest and principal metrics average repetition-specific values. They must not be interpreted as one common confusion matrix. EASI75 sensitivity used 49 and 54 observed responders at W12 and W24; specificity used 36 and 13 observed nonresponders, respectively, in both repetitions. Patient-bootstrap intervals for sensitivity, specificity and overall agreement are provided in `primary_EASI75_uncertainty.csv`. The empirical cumulative absolute-error curve is the average of the two repetition-specific proportions. Pointwise bootstrap intervals carry all saved predictions from each sampled patient together and do not refit the development procedure. Applying the published formulas used the published rounded coefficients without local refitting, with local IL2RA as a nominal sIL-2R proxy and identical eligible patients for both methods. Original p-EASI outputs were initially left unbounded to preserve the published arithmetic, whereas local estimates were clipped to 0–72; matched clipping of the original outputs was examined separately. Patient-paired bootstrap intervals were calculated for differences between the current and published-formula metrics, retaining both repetitions for each sampled patient. These intervals characterize the local comparison; they do not remove the difference in inputs or establish assay equivalence.[5–7]

1.5 Independent auxiliary target models

PP-NRS and adult DLQI were fitted separately with original-scale or $\log(1+x)$ ridge regression, using patient-level inner validation to select the link and penalty. Predictions were bounded to 0–10 and 0–30, respectively. Recorded IGA used a penalized proportional-odds model over the six observed grades, with ordered thresholds and six category probabilities. Normalized ranked probability score was the primary ordinal loss. All three targets used the same four-input panel for a practical shared measurement set, with BSA-only, molecular-only and training-only constant comparators. Neither EASI nor the target's current observed score was an input. Development and temporal application followed the patient-level W0/W12 and W24 separation used above. This limited shared-panel study does not claim to select the optimal molecular set independently for each target. Details and full estimates are in the supplement.

1.6 Reproducibility and reporting

The model analysis used Python 3.9.6, NumPy 2.0.2, pandas 2.3.3, SciPy 1.13.1 and scikit-learn 1.6.1. Exact dependency versions, analysis specifications, input/output hashes and execution logs accompany the code. Clinical and prediction reporting follow the relevant STROBE and TRIPOD+AI domains; applicability and risk-of-bias issues are recorded explicitly.[8–10] No non-significant difference, high correlation or overlapping mean trajectory was taken as proof of individual interchangeability.

Supplementary tables

N is the endpoint-specific evaluable denominator. Unknown outcomes are excluded from the observed rate and retained in missing-outcome bounds. Adult DLQI requires age at least 18. This table covers the specified visits through W24; exploratory W48 component rates remain in the CSV collection. Exact tests and correction families accompany `endpoint_tests.csv` and `endpoint_weighted_contrasts.csv`.

Table S1: Exact target attainment, observed denominators and Wilson 95% confidence intervals.

Week and target	Initial treatment	Achieved/N	Rate, %	95% CI
W0: Adult DLQI at most 10	A100	16/32	50.0	33.6–66.4
W0: Adult DLQI at most 10	A200	9/34	26.5	14.6–43.1
W0: Adult DLQI at most 10	D	12/47	25.5	15.3–39.5
W0: Adult DLQI at most 4	A100	3/32	9.4	3.2–24.2
W0: Adult DLQI at most 4	A200	0/34	0.0	0.0–10.2
W0: Adult DLQI at most 4	D	4/47	8.5	3.4–19.9
W0: EASI at most 3	A100	4/56	7.1	2.8–17.0
W0: EASI at most 3	A200	1/42	2.4	0.4–12.3
W0: EASI at most 3	D	0/65	0.0	0.0–5.6
W0: EASI at most 5	A100	9/56	16.1	8.7–27.8
W0: EASI at most 5	A200	3/42	7.1	2.5–19.0
W0: EASI at most 5	D	5/65	7.7	3.3–16.8
W0: IGA 0/1	A100	0/56	0.0	0.0–6.4
W0: IGA 0/1	A200	0/42	0.0	0.0–8.4
W0: IGA 0/1	D	0/65	0.0	0.0–5.6
W0: PP-NRS 0/1	A100	3/55	5.5	1.9–14.9
W0: PP-NRS 0/1	A200	1/42	2.4	0.4–12.3
W0: PP-NRS 0/1	D	0/65	0.0	0.0–5.6
W0: PP-NRS at most 4	A100	11/55	20.0	11.6–32.4
W0: PP-NRS at most 4	A200	7/42	16.7	8.3–30.6
W0: PP-NRS at most 4	D	5/65	7.7	3.3–16.8
W2: PP-NRS at most 4	A100	34/37	91.9	78.7–97.2
W2: PP-NRS at most 4	A200	33/36	91.7	78.2–97.1
W2: PP-NRS at most 4	D	15/27	55.6	37.3–72.4
W4: Adult DLQI at most 10	A100	14/16	87.5	64.0–96.5
W4: Adult DLQI at most 10	A200	13/15	86.7	62.1–96.3
W4: Adult DLQI at most 10	D	21/26	80.8	62.1–91.5
W4: Adult DLQI at most 4	A100	12/16	75.0	50.5–89.8
W4: Adult DLQI at most 4	A200	8/15	53.3	30.1–75.2
W4: Adult DLQI at most 4	D	11/26	42.3	25.5–61.1
W4: EASI50	A100	26/34	76.5	60.0–87.6
W4: EASI50	A200	14/18	77.8	54.8–91.0
W4: EASI50	D	14/39	35.9	22.7–51.6
W4: EASI75	A100	15/34	44.1	28.9–60.5
W4: EASI75	A200	5/18	27.8	12.5–50.9
W4: EASI75	D	6/39	15.4	7.2–29.7
W4: EASI90	A100	10/34	29.4	16.8–46.2
W4: EASI90	A200	3/18	16.7	5.8–39.2
W4: EASI90	D	0/39	0.0	0.0–9.0
W4: EASI at most 3	A100	21/34	61.8	45.0–76.1
W4: EASI at most 3	A200	4/18	22.2	9.0–45.2
W4: EASI at most 3	D	4/39	10.3	4.1–23.6
W4: EASI at most 5	A100	24/34	70.6	53.8–83.2
W4: EASI at most 5	A200	10/18	55.6	33.7–75.4
W4: EASI at most 5	D	9/39	23.1	12.6–38.3
W4: IGA 0/1	A100	13/34	38.2	23.9–55.0
W4: IGA 0/1	A200	2/17	11.8	3.3–34.3
W4: IGA 0/1	D	0/39	0.0	0.0–9.0
W4: IGA decrease at least 2	A100	18/34	52.9	36.7–68.5
W4: IGA decrease at least 2	A200	8/17	47.1	26.2–69.0
W4: IGA decrease at least 2	D	8/39	20.5	10.8–35.5
W4: PP-NRS 0/1	A100	12/34	35.3	21.5–52.1
W4: PP-NRS 0/1	A200	11/18	61.1	38.6–79.7
W4: PP-NRS 0/1	D	6/38	15.8	7.4–30.4
W4: PP-NRS decrease at least 4	A100	24/29	82.8	65.5–92.4

Week and target	Initial treatment	Achieved/N	Rate, %	95% CI
W4: PP-NRS decrease at least 4	A200	14/16	87.5	64.0–96.5
W4: PP-NRS decrease at least 4	D	22/36	61.1	44.9–75.2
W4: PP-NRS at most 4	A100	33/34	97.1	85.1–99.5
W4: PP-NRS at most 4	A200	16/18	88.9	67.2–96.9
W4: PP-NRS at most 4	D	26/38	68.4	52.5–80.9
W4: Both domains	A100	26/34	76.5	60.0–87.6
W4: Both domains	A200	13/17	76.5	52.7–90.4
W4: Both domains	D	18/39	46.2	31.6–61.4
W4: Patient domain	A100	35/35	100.0	90.1–100.0
W4: Patient domain	A200	16/18	88.9	67.2–96.9
W4: Patient domain	D	33/37	89.2	75.3–95.7
W4: Physician domain	A100	26/34	76.5	60.0–87.6
W4: Physician domain	A200	14/17	82.4	59.0–93.8
W4: Physician domain	D	18/39	46.2	31.6–61.4
W12: Adult DLQI at most 10	A100	22/22	100.0	85.1–100.0
W12: Adult DLQI at most 10	A200	24/26	92.3	75.9–97.9
W12: Adult DLQI at most 10	D	29/34	85.3	69.9–93.6
W12: Adult DLQI at most 4	A100	18/22	81.8	61.5–92.7
W12: Adult DLQI at most 4	A200	20/26	76.9	57.9–89.0
W12: Adult DLQI at most 4	D	20/34	58.8	42.2–73.6
W12: EASI50	A100	38/42	90.5	77.9–96.2
W12: EASI50	A200	25/31	80.6	63.7–90.8
W12: EASI50	D	38/50	76.0	62.6–85.7
W12: EASI75	A100	26/42	61.9	46.8–75.0
W12: EASI75	A200	22/31	71.0	53.4–83.9
W12: EASI75	D	23/50	46.0	33.0–59.6
W12: EASI90	A100	19/42	45.2	31.2–60.1
W12: EASI90	A200	7/31	22.6	11.4–39.8
W12: EASI90	D	14/50	28.0	17.5–41.7
W12: EASI at most 3	A100	29/42	69.0	54.0–80.9
W12: EASI at most 3	A200	17/31	54.8	37.8–70.8
W12: EASI at most 3	D	23/50	46.0	33.0–59.6
W12: EASI at most 5	A100	34/42	81.0	66.7–90.0
W12: EASI at most 5	A200	22/31	71.0	53.4–83.9
W12: EASI at most 5	D	31/50	62.0	48.2–74.1
W12: IGA 0/1	A100	24/42	57.1	42.2–70.9
W12: IGA 0/1	A200	9/31	29.0	16.1–46.6
W12: IGA 0/1	D	10/50	20.0	11.2–33.0
W12: IGA decrease at least 2	A100	26/42	61.9	46.8–75.0
W12: IGA decrease at least 2	A200	23/31	74.2	56.8–86.3
W12: IGA decrease at least 2	D	25/50	50.0	36.6–63.4
W12: PP-NRS 0/1	A100	25/42	59.5	44.5–73.0
W12: PP-NRS 0/1	A200	24/31	77.4	60.2–88.6
W12: PP-NRS 0/1	D	23/50	46.0	33.0–59.6
W12: PP-NRS decrease at least 4	A100	28/36	77.8	61.9–88.3
W12: PP-NRS decrease at least 4	A200	27/28	96.4	82.3–99.4
W12: PP-NRS decrease at least 4	D	39/48	81.2	68.1–89.8
W12: PP-NRS at most 4	A100	37/42	88.1	75.0–94.8
W12: PP-NRS at most 4	A200	29/31	93.5	79.3–98.2
W12: PP-NRS at most 4	D	43/50	86.0	73.8–93.0
W12: Both domains	A100	21/34	61.8	45.0–76.1
W12: Both domains	A200	19/30	63.3	45.5–78.1
W12: Both domains	D	20/48	41.7	28.8–55.7
W12: Patient domain	A100	27/30	90.0	74.4–96.5
W12: Patient domain	A200	27/30	90.0	74.4–96.5
W12: Patient domain	D	29/41	70.7	55.5–82.4
W12: Physician domain	A100	30/42	71.4	56.4–82.8
W12: Physician domain	A200	22/31	71.0	53.4–83.9
W12: Physician domain	D	26/50	52.0	38.5–65.2
W24: Adult DLQI at most 10	A100	12/12	100.0	75.8–100.0
W24: Adult DLQI at most 10	A200	16/18	88.9	67.2–96.9
W24: Adult DLQI at most 10	D	22/24	91.7	74.2–97.7
W24: Adult DLQI at most 4	A100	11/12	91.7	64.6–98.5
W24: Adult DLQI at most 4	A200	16/18	88.9	67.2–96.9
W24: Adult DLQI at most 4	D	20/24	83.3	64.1–93.3
W24: EASI50	A100	29/29	100.0	88.3–100.0

Week and target	Initial treatment	Achieved/N	Rate, %	95% CI
W24: EASI50	A200	21/24	87.5	69.0–95.7
W24: EASI50	D	35/40	87.5	73.9–94.5
W24: EASI75	A100	26/29	89.7	73.6–96.4
W24: EASI75	A200	21/24	87.5	69.0–95.7
W24: EASI75	D	30/40	75.0	59.8–85.8
W24: EASI90	A100	19/29	65.5	47.3–80.1
W24: EASI90	A200	14/24	58.3	38.8–75.5
W24: EASI90	D	16/40	40.0	26.3–55.4
W24: EASI at most 3	A100	23/29	79.3	61.6–90.2
W24: EASI at most 3	A200	17/24	70.8	50.8–85.1
W24: EASI at most 3	D	22/40	55.0	39.8–69.3
W24: EASI at most 5	A100	27/29	93.1	78.0–98.1
W24: EASI at most 5	A200	21/24	87.5	69.0–95.7
W24: EASI at most 5	D	28/40	70.0	54.6–81.9
W24: IGA 0/1	A100	21/29	72.4	54.3–85.3
W24: IGA 0/1	A200	15/23	65.2	44.9–81.2
W24: IGA 0/1	D	14/40	35.0	22.1–50.5
W24: IGA decrease at least 2	A100	24/29	82.8	65.5–92.4
W24: IGA decrease at least 2	A200	20/23	87.0	67.9–95.5
W24: IGA decrease at least 2	D	27/40	67.5	52.0–79.9
W24: PP-NRS 0/1	A100	22/29	75.9	57.9–87.8
W24: PP-NRS 0/1	A200	19/24	79.2	59.5–90.8
W24: PP-NRS 0/1	D	23/41	56.1	41.0–70.1
W24: PP-NRS decrease at least 4	A100	21/25	84.0	65.3–93.6
W24: PP-NRS decrease at least 4	A200	21/22	95.5	78.2–99.2
W24: PP-NRS decrease at least 4	D	31/38	81.6	66.6–90.8
W24: PP-NRS at most 4	A100	29/29	100.0	88.3–100.0
W24: PP-NRS at most 4	A200	23/24	95.8	79.8–99.3
W24: PP-NRS at most 4	D	37/41	90.2	77.5–96.1
W24: Both domains	A100	22/24	91.7	74.2–97.7
W24: Both domains	A200	20/23	87.0	67.9–95.5
W24: Both domains	D	26/36	72.2	56.0–84.2
W24: Patient domain	A100	23/24	95.8	79.8–99.3
W24: Patient domain	A200	20/22	90.9	72.2–97.5
W24: Patient domain	D	31/35	88.6	74.0–95.5
W24: Physician domain	A100	28/29	96.6	82.8–99.4
W24: Physician domain	A200	22/24	91.7	74.2–97.7
W24: Physician domain	D	32/40	80.0	65.2–89.5

Table S2: Primary performance of the common-model development procedure and matched controls.

	W0	W12	W24
Patients	152	89	75
MAE (95% CI)	3.19 (2.75, 3.70)	1.30 (0.96, 1.71)	1.19 (0.89, 1.53)
BSA-only area MAE	3.12	1.18	1.02
Three-marker-only MAE	7.63	7.57	8.69
Primary minus BSA-only area			
MAE difference (95% CI)	0.07 (−0.03, 0.17)	0.12 (0.04, 0.20)	0.17 (0.11, 0.24)
RMSE	4.45	2.29	1.87
Mean signed error	−0.59 (−1.27, 0.07)	0.82 (0.43, 1.29)	0.84 (0.50, 1.25)
Spearman correlation	0.918	0.927	0.904
Concordance correlation	0.910	0.900	0.869
R^2	0.819	0.770	0.656
Calibration intercept	2.01 (0.92, 3.28)	0.13 (−0.32, 0.69)	0.07 (−0.20, 0.42)
Calibration slope	0.91 (0.82, 1.00)	0.81 (0.66, 0.94)	0.73 (0.59, 0.82)
Within one point, %	24.0	57.9	59.3
Within two points, %	41.8	84.3	82.0
Severity-band agreement, %	83.2	88.2	92.7
Sensitivity for EASI >7, %	92.5	85.3	100.0
Specificity for EASI ≤7, %	92.1	90.3	91.9

W0/W12 use patient-grouped nested validation; W24 is same-cohort temporal application. Structure and penalty selection are internal to each training fold; earlier marker-panel selection is not repeated. Matched controls use the same patients and outer folds. BSA-only area uses the zero-intercept area structure; three-marker-only uses CCL17, CCL26 and IL-5 without BSA. Positive paired MAE differences mean greater primary-procedure error. Metrics are averaged after scoring each repetition. Intervals are 95% patient-bootstrap intervals conditional on saved predictions; selection and fitting are not repeated. MAE, mean absolute error; RMSE, root mean squared error. Positive signed error indicates overestimation. Severity-band agreement uses four EASI bands for observed and predicted values: ≤ 7, > 7 to ≤ 21, > 21 to ≤ 50, and > 50. Calibration regresses observed EASI on predicted EASI; ideal intercept/slope are 0/1. The single final formula is not independently validated.

Table S3: Coverage of recorded safety documentation by follow-up window.

Initial treatment	Initial N	With note through W24, n/N	No note through W24, n	With note at any follow-up, n/N	Note records through W24, n
Dupilumab	65	27/65	38	28/65	30
Abrocitinib 100 mg	56	20/56	36	20/56	28
Abrocitinib 200 mg	42	19/42	23	19/42	37

Patients are counted once within each documentation window. The through-W24 window uses recorded nominal visits up to W24; the all-follow-up column also includes later recorded visits. Note records can repeat within a patient and are not adverse-event counts. Absence of a note means unknown documentation, not absence of an event. Initial-group N is a cohort denominator, not a verified safety-observation denominator. These counts do not estimate AE or SAE incidence, severity or causality, and do not establish event-time treatment attribution. Source, `safety_documentation_windows.csv`.

Table S4: Robustness to training-only exclusion of reference-inconsistent development visits.

	W0	W12	W24
All original held-out observations			
Patients	152	89	75
Primary MAE (95% CI)	3.19 (2.75, 3.70)	1.30 (0.96, 1.71)	1.19 (0.89, 1.53)
Training-exclusion MAE (95% CI)	3.19 (2.72, 3.70)	1.15 (0.82, 1.57)	1.06 (0.75, 1.40)
Paired ΔMAE (95% CI)	−0.007 (−0.090, 0.078)	−0.149 (−0.228, −0.073)	−0.134 (−0.206, −0.064)
Primary calibration slope	0.913	0.814	0.735
Training-exclusion calibration slope	0.915	0.821	0.733
Paired Δslope (95% CI)	0.002 (−0.008, 0.015)	0.007 (−0.004, 0.028)	−0.002 (−0.015, 0.014)
Strict reference-consistent evaluation subset			
Patients	140	86	71
Primary MAE (95% CI)	3.17 (2.70, 3.71)	1.32 (0.97, 1.75)	1.11 (0.82, 1.42)
Training-exclusion MAE (95% CI)	3.19 (2.72, 3.72)	1.18 (0.83, 1.60)	0.97 (0.70, 1.30)
Paired ΔMAE (95% CI)	0.012 (−0.073, 0.101)	−0.143 (−0.223, −0.066)	−0.131 (−0.203, −0.063)
Primary calibration slope	0.910	0.790	0.765
Training-exclusion calibration slope	0.910	0.798	0.766
Paired Δslope (95% CI)	0.001 (−0.010, 0.014)	0.008 (−0.003, 0.035)	0.001 (−0.013, 0.018)

Eight W0 and two W12 visits with complete four-region scores whose sum differed from recorded total EASI by more than 0.15 points were excluded from development fitting only. Visits with missing regional references were retained. Original patient outer folds, evaluation patients and recorded outcomes were unchanged; preprocessing and inner structure/penalty selection were repeated. The strict evaluation subset requires all four regions and an absolute sum-to-total difference at most 0.01. Paired Δ is training-exclusion minus primary; negative ΔMAE means less error. Calibration regresses observed on predicted EASI, with ideal slope 1. Metrics are scored within each repetition before averaging. Intervals use 2,000 paired patient resamples conditional on the saved predictions (seed 20261002); bootstrap resamples do not repeat fitting or selection. This is robustness to exclusion, not correction of clinical truth, independent validation or a replacement of the primary formula. Matched refitted BSA controls and complete metrics are in the `reference_sensitivity` aggregate directory.

Table S5: Input and clinical-reference specifications, with unresolved source requirements.

Element	Current analysis specification	Evidence boundary
EASI reference	Recorded total EASI, 0–72; observed outcomes only, with no imputed validation truth.	Regional-sum discrepancies are audited; the authoritative field in discrepant records and blinded repeat-assessment reliability are not confirmed.
BSA input	Affected BSA entered from 0 to 100 (10% is entered as 10). It supplies a clinical extent component of the hybrid estimate.	The source does not establish an observer-independent measurement protocol; this is not a blood-only score.
Molecular inputs	CCL17, CCL26 and IL-5. Positive finite integration-export numerics recorded in pg/mL are log2 transformed once.	A positive numeric export is not proof of validated quantification. Transfer to another export or assay requires confirmed scale compatibility.
Specimen and assay	Laboratory reports describe serum, Luminex LXSAHM-13/Luminex-200 and MSD K15067M-1/K15067M-2/SQ120 assays.	Matched specimen handling, technical-repeat precision and cross-platform bridging are not established by the analysis.
Dilution and limits	Reports describe 50-microlitre serum and twofold laboratory dilution. No additional dilution factor is applied to the integration export.	Whether dilution is already corrected in that export, and validated lower/upper quantification limits, require source confirmation. Software calibration boundaries are not validated LLOQ/ULOQ.
Clinical/serum timing	Patient identifier plus nominal W0, W12 or W24 links the clinical and molecular record.	Exact calendar-day separation and a permitted sampling window are unverified; nominal matching does not establish same-day measurement.
Development scope	Age $\geq$12, observed EASI/BSA and all three positive molecular inputs. Fitting and selection use W0/W12 with patient-grouped inner and outer validation.	The marker panel arose from earlier exploration. W24 had previously been inspected and is same-cohort temporal application, not untouched external validation.
Calculator output	The precise frozen coefficients return EASI units clipped to 0–72. Missing, nonfinite or nonpositive molecular inputs and invalid BSA are rejected.	BSA=0 returns zero by the area constraint; it does not prove absence of molecular inflammation. Independent validation of the single final coefficient vector is unavailable.
Recorded IGA	Recorded grades 0–5 are retained. IGA 0/1 and decrease $\geq$2 use those recorded values without conversion to a 0–4 scale.	Exact instrument/version and correspondence to a validated IGA system require source confirmation.
PP-NRS and DLQI	Recorded PP-NRS uses 0–10; adult DLQI uses 0–30 and age $\geq$18. A PP-NRS decrease $\geq$4 requires observed baseline $\geq$4 and observed follow-up.	PP-NRS recall window and exact questionnaire versions remain unconfirmed. Adult DLQI is unknown in adolescent composite rules.
IgE and eosinophils	Analyses retain the recorded numeric values; adjusted models use log(1+x) and back-transformation.	Clinical laboratory units and the meaning of repeated IgE values of 5000 require source confirmation; no censoring threshold is inferred from repetition alone.
Treatment targets	The September investigator framework supplies stage-specific physician and patient thresholds, with the stated three-valued missing-component logic.	The exact approved and prospectively implemented framework version and original voting records are not verified; target attainment is not validation of a new consensus or management effect.

Independent auxiliary target models

Auxiliary clinical targets. We fitted separate models for PP-NRS, adult DLQI and the recorded six-category IGA using the same restricted inputs as the EASI model: BSA, CCL17, CCL26 and IL-5. Continuous targets used ridge regression on the original or log-transformed scale, selected by patient-level inner validation and bounded to the scale range. Recorded IGA used a proportional-odds ordinal logistic model, selected by normalized ranked probability score (RPS). All fits used observed baseline and week-12 outcomes. Week 24 was excluded from model selection. DLQI analyses were restricted to participants aged at least 18 years. These models estimated separate targets; the EASI coefficients were not reused as their prediction formulas.

The common input panel did not yield comparably accurate estimates of itch or quality of life. PP-NRS MAE was 2.35, 2.78 and 2.53 points at baseline, week 12 and week 24, respectively (151, 89 and 75 participants). Adult DLQI MAE was 6.25, 5.90 and 6.28 points (105, 58 and 44 participants). For recorded IGA, normalized RPS was 0.071, 0.099 and 0.124, with exact-category agreement of 54.3%, 35.4% and 28.0% (152, 89 and 75 participants). Adding the three molecular measurements to BSA did not produce a consistent improvement for PP-NRS or DLQI, and ordinal performance was poorer than the BSA-only reference at weeks 12 and 24. These findings limit the interpretation of the common panel to the outcomes for which it was informative. They do not establish that every independently selected panel would fail. The exact identity of the recorded 0–5 IGA instrument requires confirmation, so these ordinal results are reported as exploratory analyses of recorded grades.

Transport of published equations

Published p-EASI coefficients were applied unchanged to identical eligible records, retaining both raw and 0–72-clipped outputs. The numerical source tables distinguish the primary nested procedure from fixed-structure, BSA-only and molecular-only comparisons. Published p-EASI uses serum alone, whereas the present procedure includes clinical BSA; local IL2RA is an unbridged proxy for sIL-2R. These local applications assess transport under unverified assay compatibility and do not establish model superiority or an ethnicity-specific difference.[5–7]

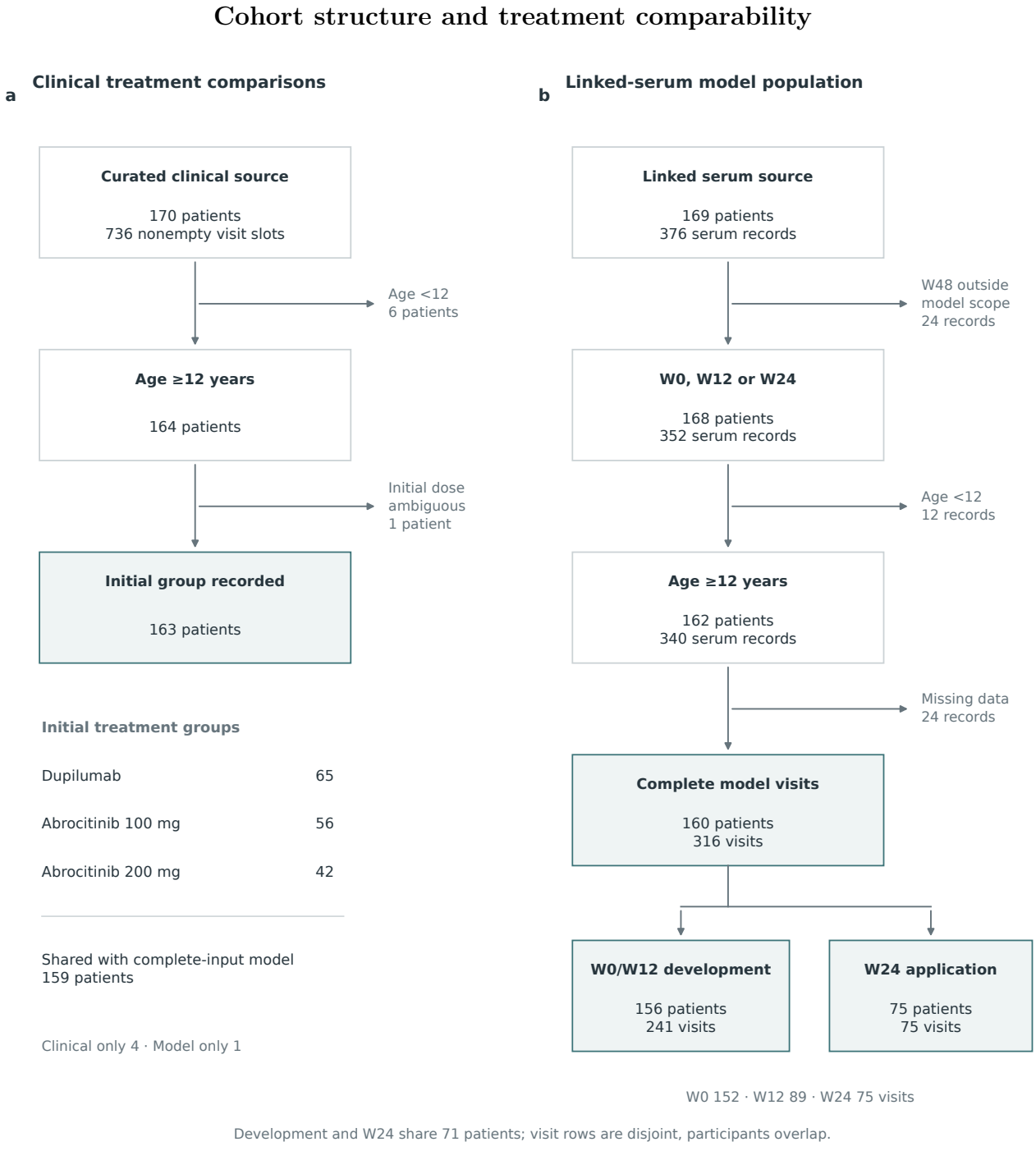


Figure S37: **Parallel clinical and complete-input model populations.** a, Clinical comparison eligibility starts with 170 curated patients and 1,700 predefined visit slots; 736 slots contain recorded visit data. Six patients aged <12 and one age-eligible patient with an ambiguous initial dose are excluded, leaving 163 patients. This is a record source, not a screened or consented recruitment denominator. b, The linked serum branch is separate and is not a subset of the clinical comparison. Serum records count linked sample rows, not independent participants. Excluding 24 W48 records leaves 168 patients and 352 W0/W12/W24 records; age exclusions remove 12 records from six patients. Complete-case exclusions are sequential and mutually exclusive at the record level. Of the remaining 340 records, 13 lack EASI, one BSA, none CCL17, eight CCL26 and two IL-5, leaving 316 visits from 160 patients. Missing model inputs include unavailable valid positive molecular values. Losing a visit does not necessarily remove the patient. Development comprises 156 patients and 241 visits, W0 n=152 and W12 n=89. W24 comprises 75 patients and 75 visits and is previously inspected within-cohort temporal application, not external validation. The clinical and model populations share 159 patients, with four clinical-only and one model-only. Development and W24 share 71 patients, with 85 development-only and four W24-only. Fifty-one patients have all three complete model visits. Initial-dose ambiguity may remain in the treatment-free model. No missing clinical outcome is filled as validation truth. All exact counts and units remain in the linked flow and overlap source tables.

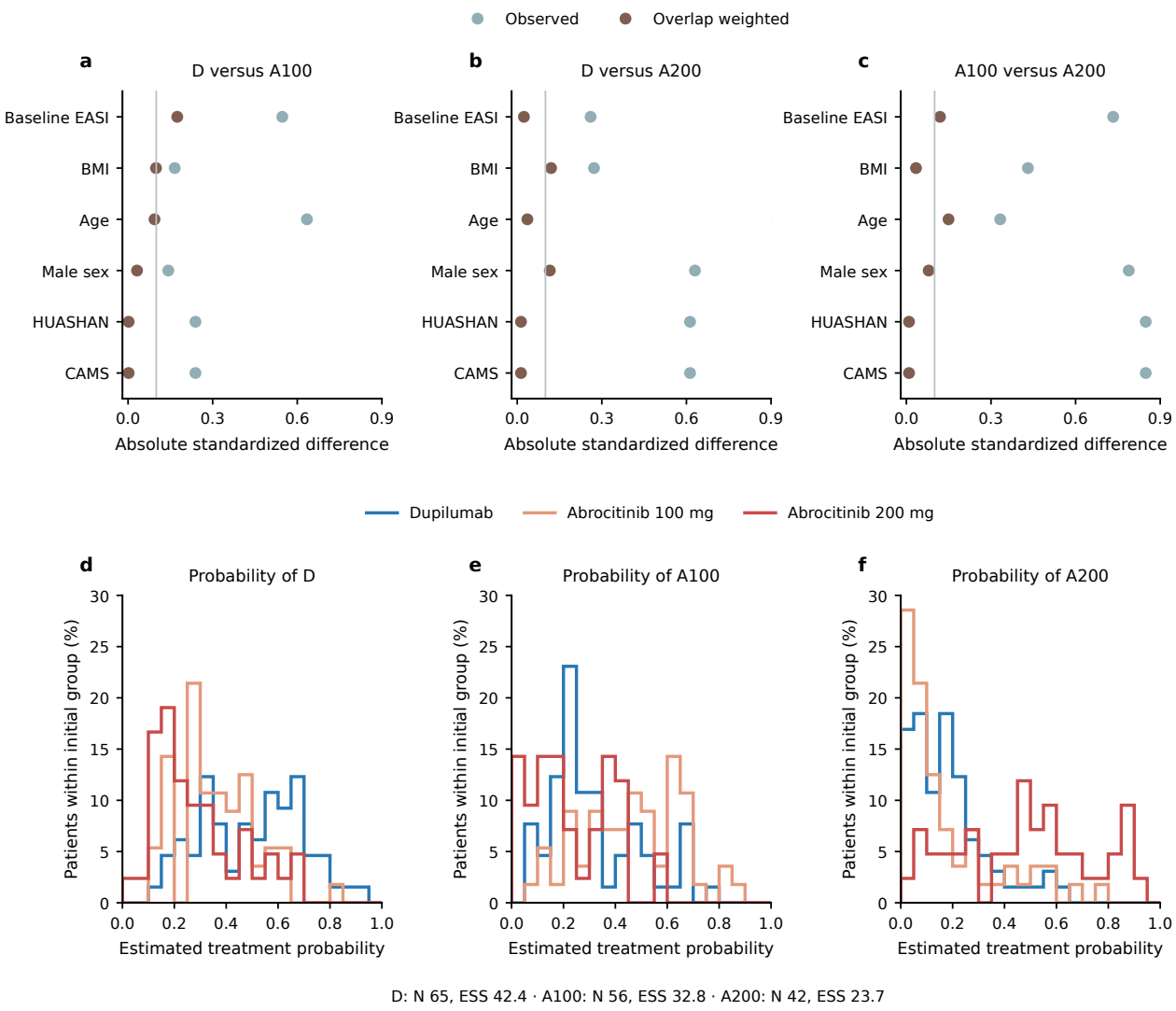


Figure S38: **Baseline balance, treatment overlap and effective sample size.** a–c, Absolute standardized mean differences for every covariate in the initial-treatment propensity model, before and after generalized overlap weighting. Differences use the unweighted pairwise pooled standard deviation in both displays. The light 0.10 reference line is a descriptive balance guide, not a statistical significance threshold. Residual imbalance remains. d–f, Baseline multinomial treatment probabilities, stratified by actual initial treatment; each distribution uses all patients in that initial group and sums to 100%. These show common support rather than longitudinal drug exposure. The footer reports original N and effective sample size, $(\text{sum w})^2/\text{sum}(\text{w}^2)$. All centre, sex, age, BMI and EASI variables are shown. Overlap weighting targets comparable baseline patients; it cannot remove unmeasured confounding or correct outcome-driven dose changes. Endpoint- and marker-specific availability/ESS diagnostics are retained in the source tables.

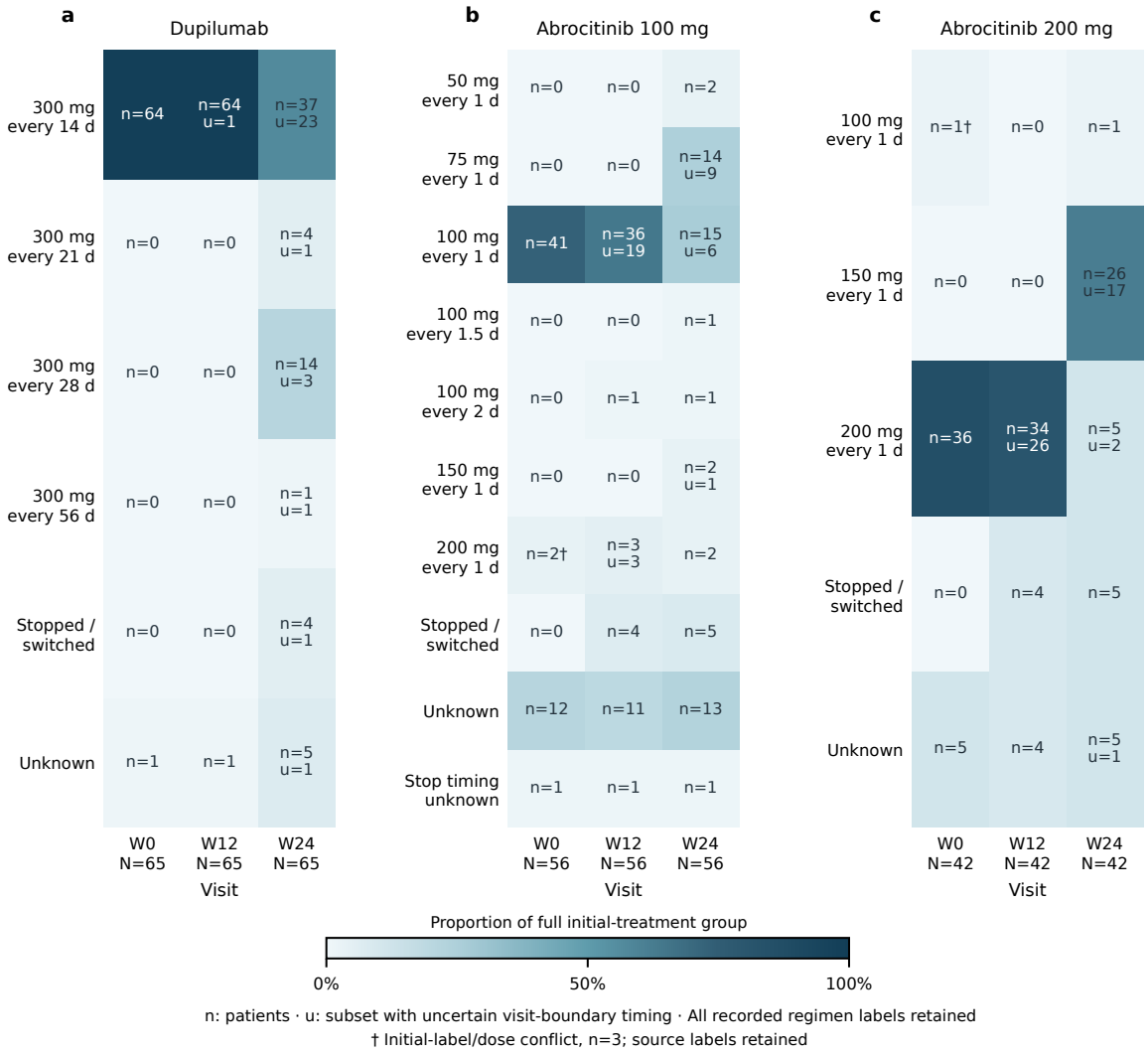


Figure S39: **Recorded regimens evolve after the initial treatment assignment.** Within each initial-treatment group, the table-like heatmap shows all recorded regimens at W0/W12/W24. Cell n values are patient counts; u denotes the subset with uncertain timing relative to the visit boundary. Blue intensity is the proportion of the full initial-treatment group, with a fixed 0–100% scale. Zero cells are genuine zero counts. Unknown regimens, uncertain stop timing and stopping/switching are retained; all 163 patients remain represented. W0 is the recorded starting regimen and must not be read as prior systemic exposure. Daggers flag three unresolved source conflicts at W0: two patients assigned to the initial A100 source group have a recorded 200 mg regimen and one assigned to A200 has a recorded 100 mg regimen. The source group and recorded regimen are both preserved; these are not presumed switches. Recorded 75 mg and 150 mg labels are preserved without assuming that they mean alternate-day 100 mg or 200 mg. This aggregate display is not an individual transition flow and does not resolve whether a treatment decision preceded or followed the outcome assessment.

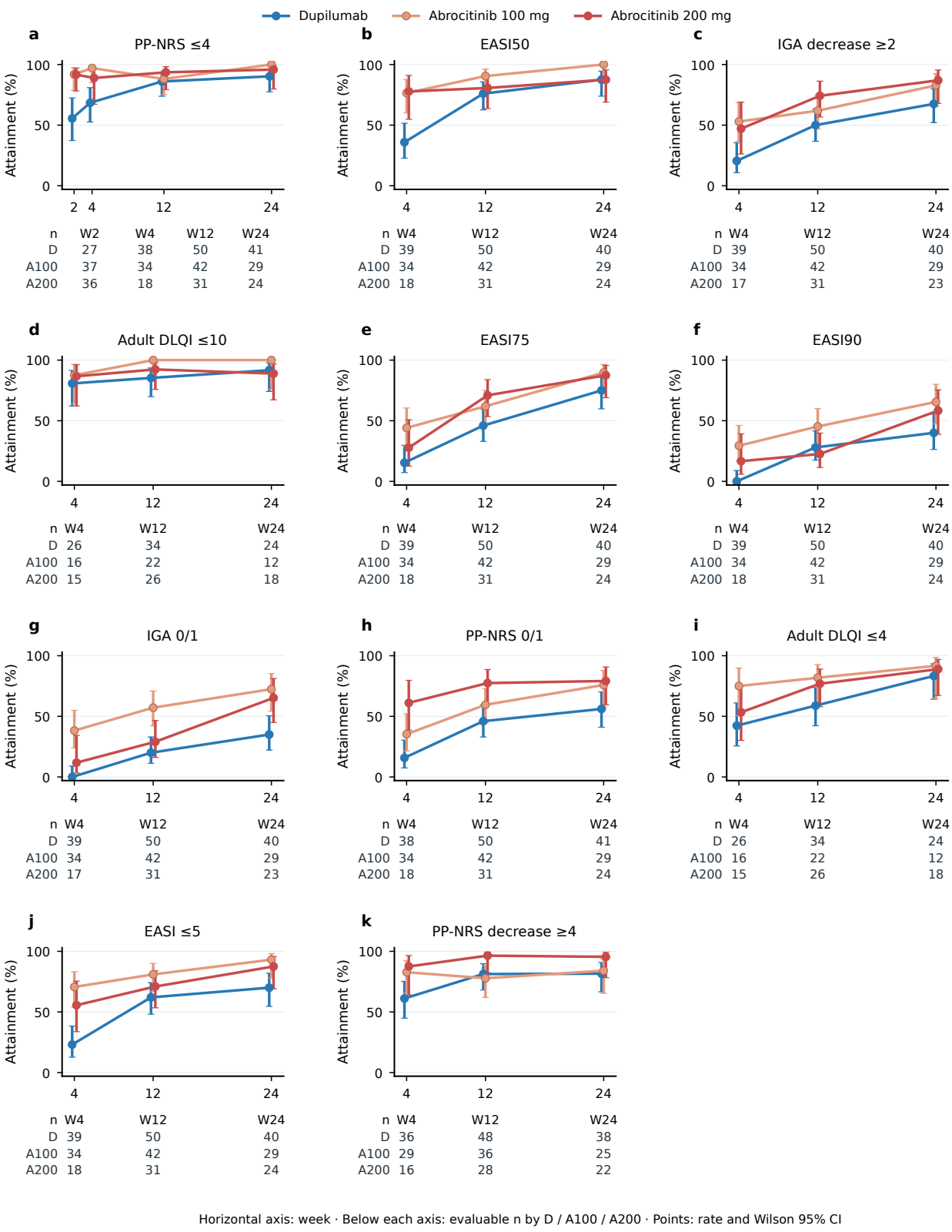


Figure S1: **Clinical response and absolute-score targets.** Observed proportions with Wilson 95% confidence intervals, by initial treatment. Count strips give the endpoint-specific evaluable denominator at each visit in D/A100/A200 order. Only PP-NRS ≤4 is evaluated at W2. The four-point PP-NRS decrease requires observed baseline PP-NRS ≥4 and an observed follow-up value. EASI percentage responses require a positive observed baseline. Adult DLQI requires age ≥18. Missing outcomes are unknown, not failures. Connecting lines describe observed visit summaries; the patient set can differ between visits. Initial treatment is retained after regimen changes.

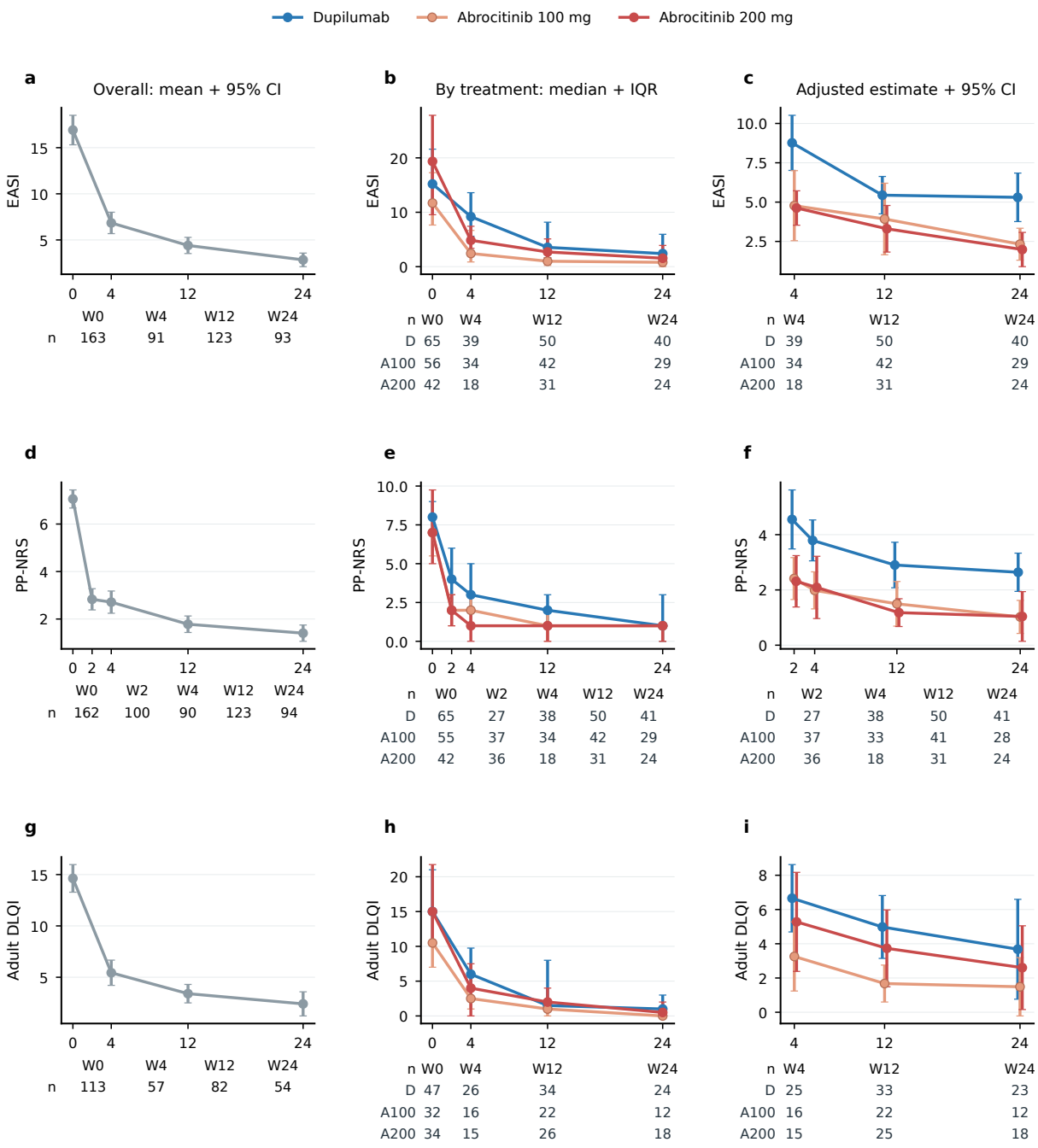


Figure S2: **Observed and baseline-adjusted EASI, PP-NRS, Adult DLQI.** Each row is one outcome. Left, overall available-case mean and 95% t confidence interval, exactly combined from group moments. Middle, initial-treatment-specific median and interquartile range. Right, overlap-weighted adjusted trajectories and patient-cluster robust 95% confidence intervals, controlling baseline outcome, age, sex, BMI and centre. Visit composition may change. Only investigator-specified visits are shown; no causal current-dose effect is claimed. Exact sample sizes and contrasts accompany the source tables.

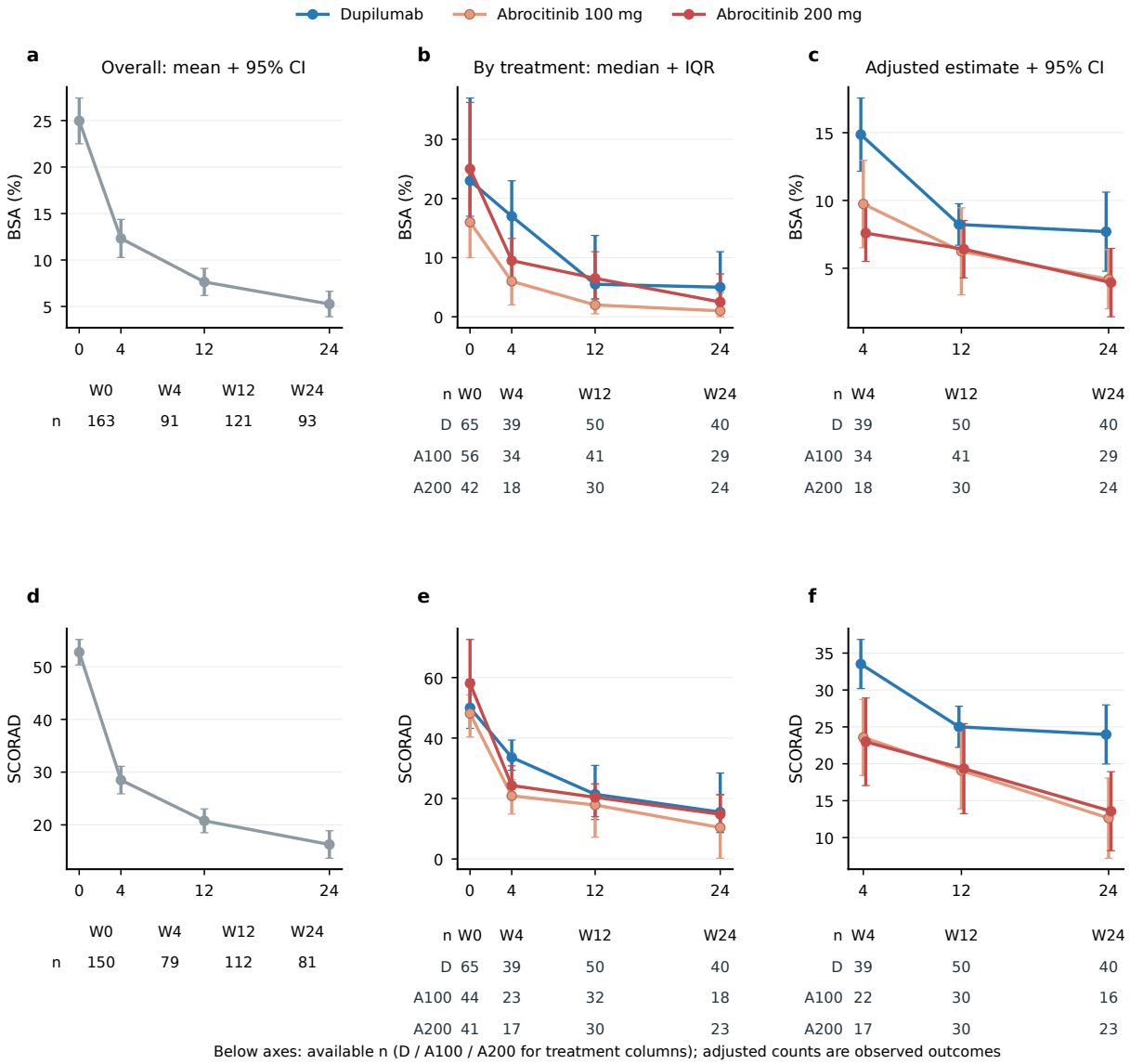


Figure S3: **Observed and baseline-adjusted BSA (%), SCORAD.** Each row is one outcome. Left, overall available-case mean and 95% t confidence interval, exactly combined from group moments. Middle, initial-treatment-specific median and interquartile range. Right, overlap-weighted adjusted trajectories and patient-cluster robust 95% confidence intervals, controlling baseline outcome, age, sex, BMI and centre. Visit composition may change. Only investigator-specified visits are shown; no causal current-dose effect is claimed. Exact sample sizes and contrasts accompany the source tables.

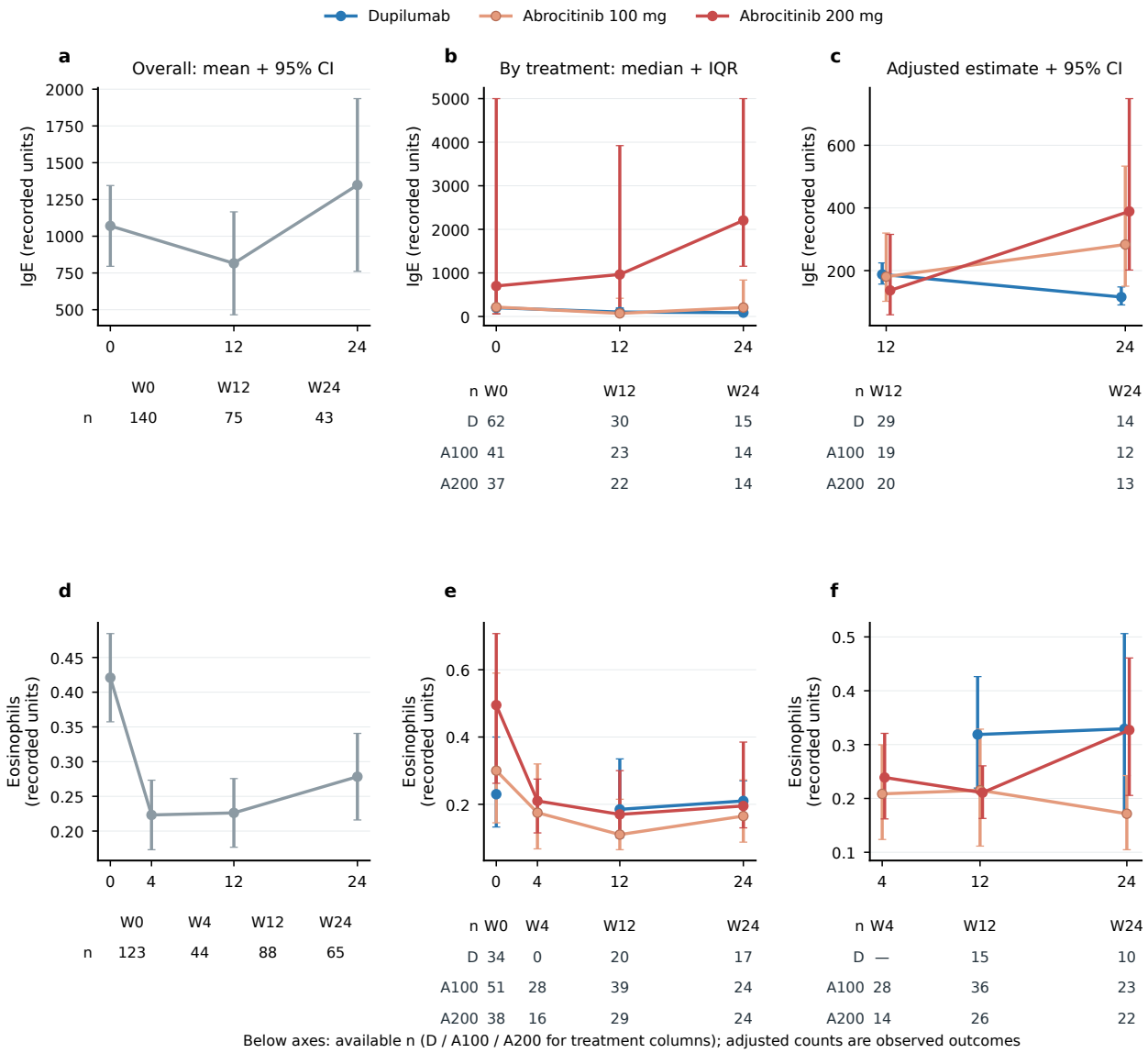


Figure S4: **Observed and baseline-adjusted IgE (recorded units), Eosinophils (recorded units).** Each row is one outcome. Left, overall available-case mean and 95% t confidence interval, exactly combined from group moments. Middle, initial-treatment-specific median and interquartile range. Right, overlap-weighted adjusted trajectories and patient-cluster robust 95% confidence intervals, controlling baseline outcome, age, sex, BMI and centre. IgE/eosinophils use log(1+x) models and back-transformation; these adjusted estimates are not arithmetic means. Visit composition may change. Only investigator-specified visits are shown; no causal current-dose effect is claimed. Exact sample sizes and contrasts accompany the source tables.

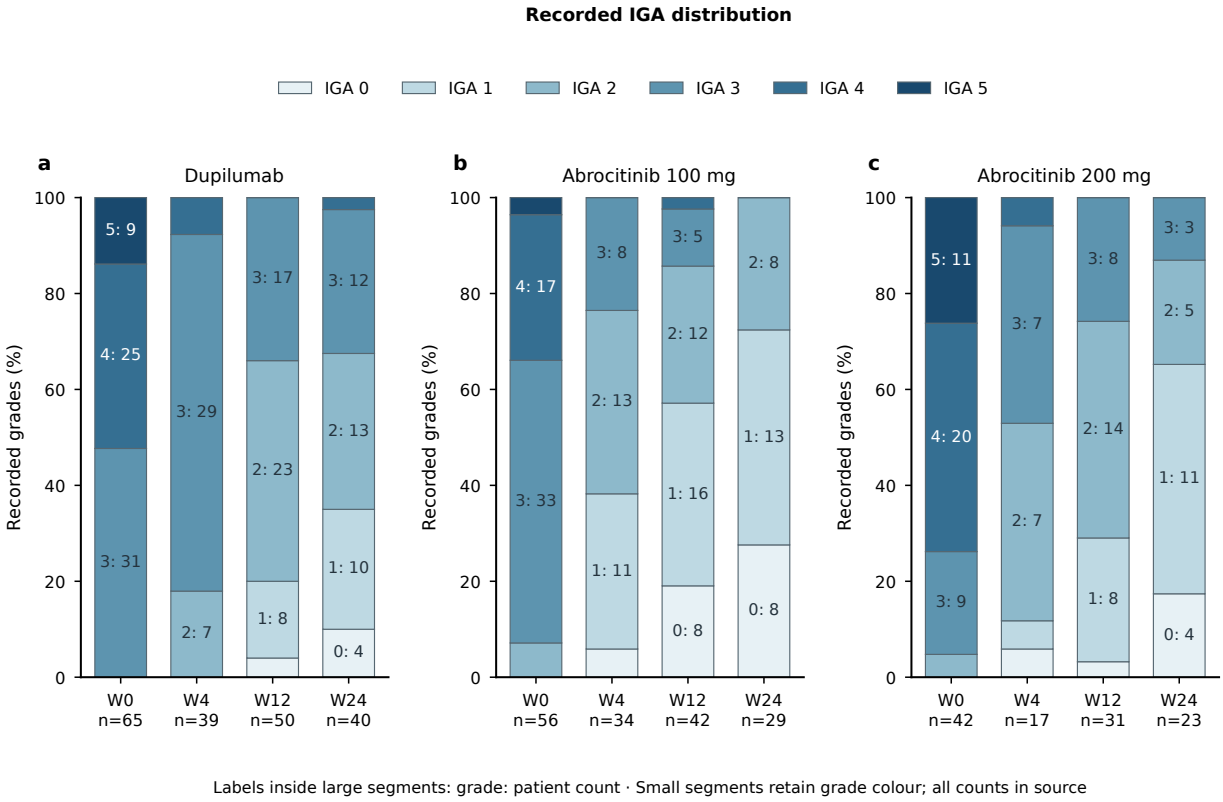


Figure S15: **Recorded IGA composition at four visits.** Each bar is the recorded IGA composition among evaluable patients, whose total n is printed below the bar. The ordered blue scale encodes recorded grade 0–5, not treatment; borders distinguish grade 0 from the background. Labels in segments occupying at least 10% show grade and exact patient count; all grade counts remain in source data. Grade 5 remains as recorded because instrument identity is unresolved; no conversion to a standard 0–4 scale was made. EASI90 is shown with the other clinical targets in Figure S1.

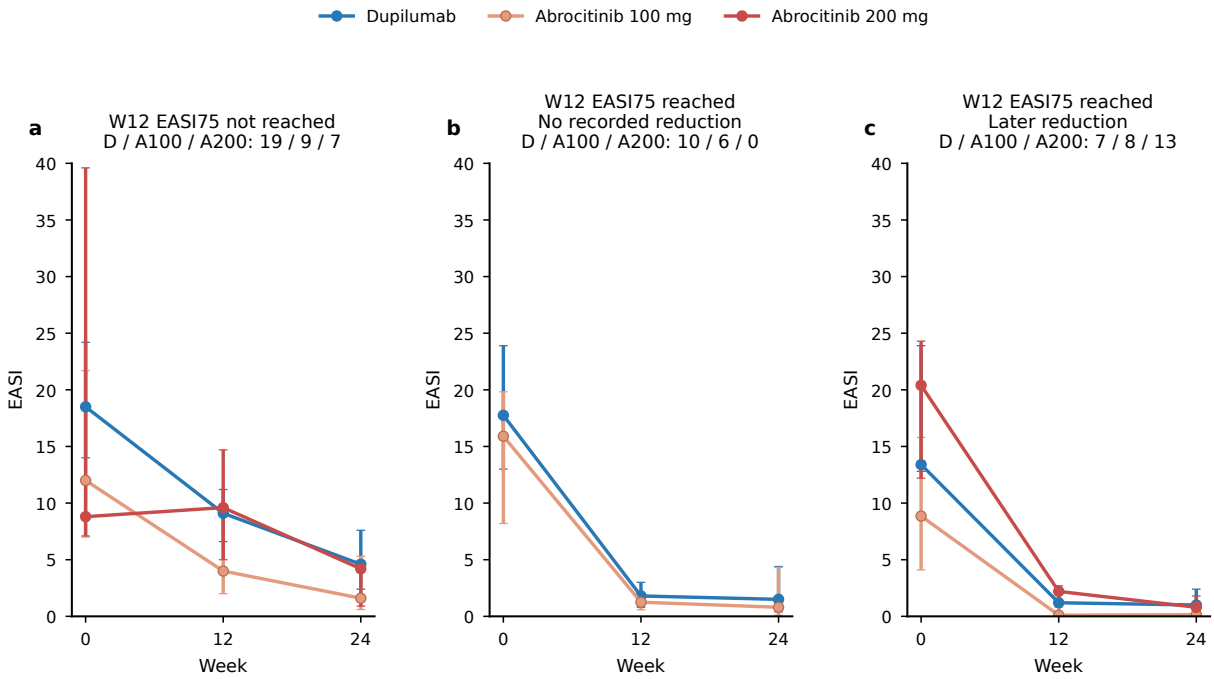


Figure S16: **EASI trajectories before and after the W12 treatment decision.** Each panel is one recorded pathway: W12 EASI75 not reached, reached without recorded reduction, or reached followed by reduction. All trajectories retain the same patients observed at W0/W12/W24; treatment is encoded by colour with solid lines and circle markers. Median and patient-bootstrap 95% confidence intervals are shown. Panel counts are ordered D/A100/A200; the zero-size A200 group in the no-reduction panel has no trajectory. Paths depend on later information and are descriptive, not randomized treatment strategies. Other paths, including unknown exposure, stopping and switching, remain in the source tables.

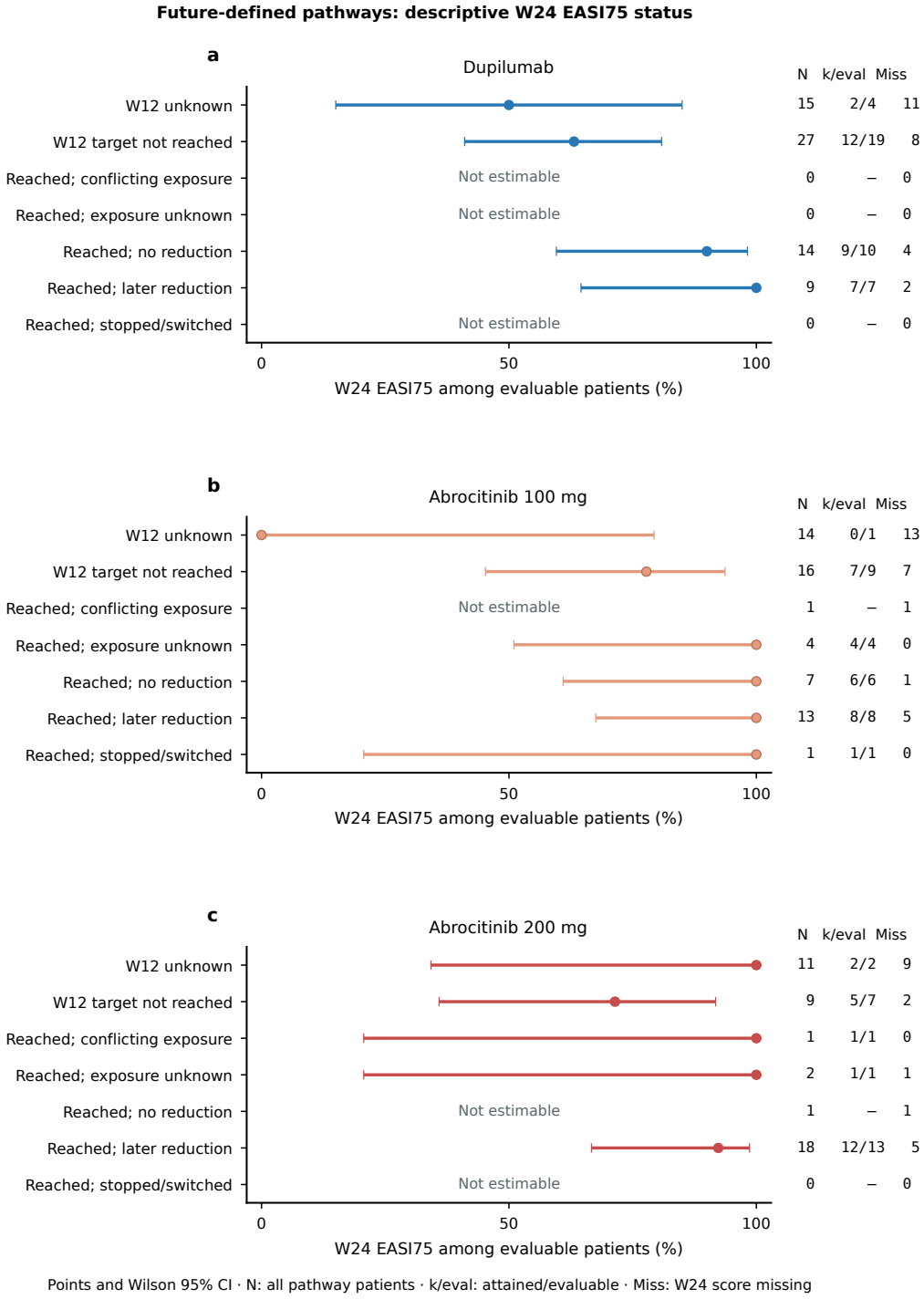


Figure S17: **W24 EASI75 status in future-defined recorded pathways.** Each row shows the observed W24 EASI75 proportion and Wilson 95% confidence interval among evaluable patients. The table states complete pathway size N, attained/evaluable k/eval and the count missing a W24 score. All recorded paths are retained, including zero-size rows, which explicitly show N=0 and Not estimable. An evaluable denominator of zero is not a zero response rate. W12 target status and later recorded treatment define these pathways; they are descriptive future-defined groups, not randomized strategies or causal tapering effects. Missing outcomes are unknown, not failures. Missing-outcome bounds remain in source data.

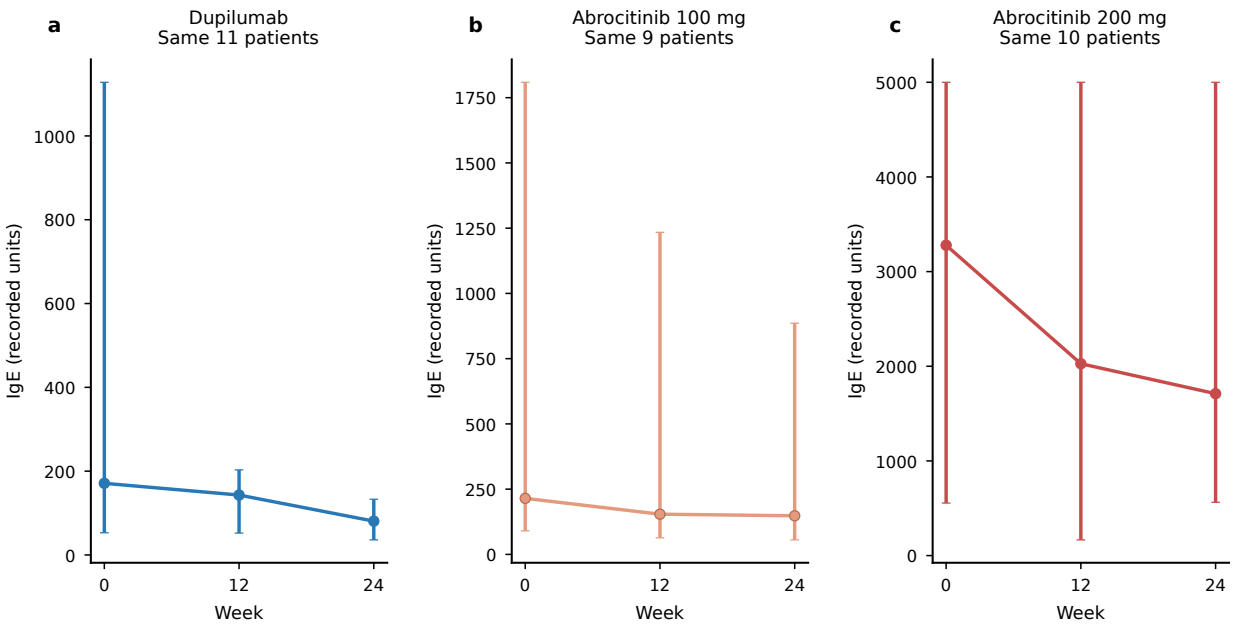


Figure S20: **IgE with the same patients retained at all three visits.** Medians and patient-bootstrap 95% confidence intervals in fixed complete W0/W12/W24 sets. Each treatment panel retains identical patients at all three visits. Vertical scales differ across treatment panels so that within-group trajectories remain visible; compare absolute values using the tick labels. These denominators differ from cross-sectional and W0–W24-only paired sets. This figure tests whether changing sample composition may explain an apparent visit-level rise; it does not establish a causal drug effect.

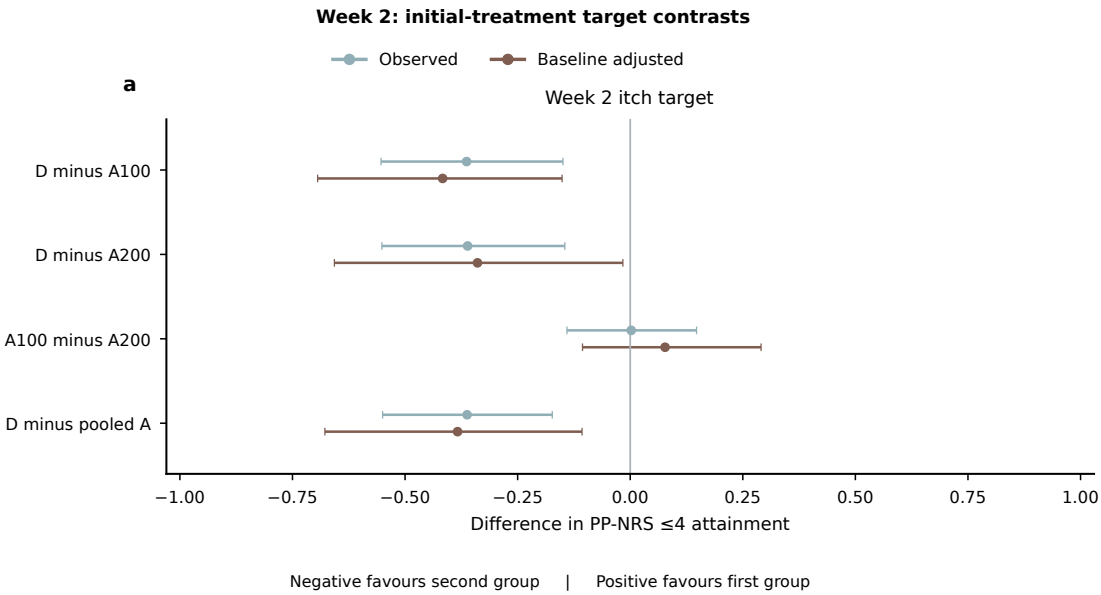


Figure S22: **Week 2: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. D denotes dupilumab; A100/A200 the initial abrocitinib doses. Pooled abrocitinib is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

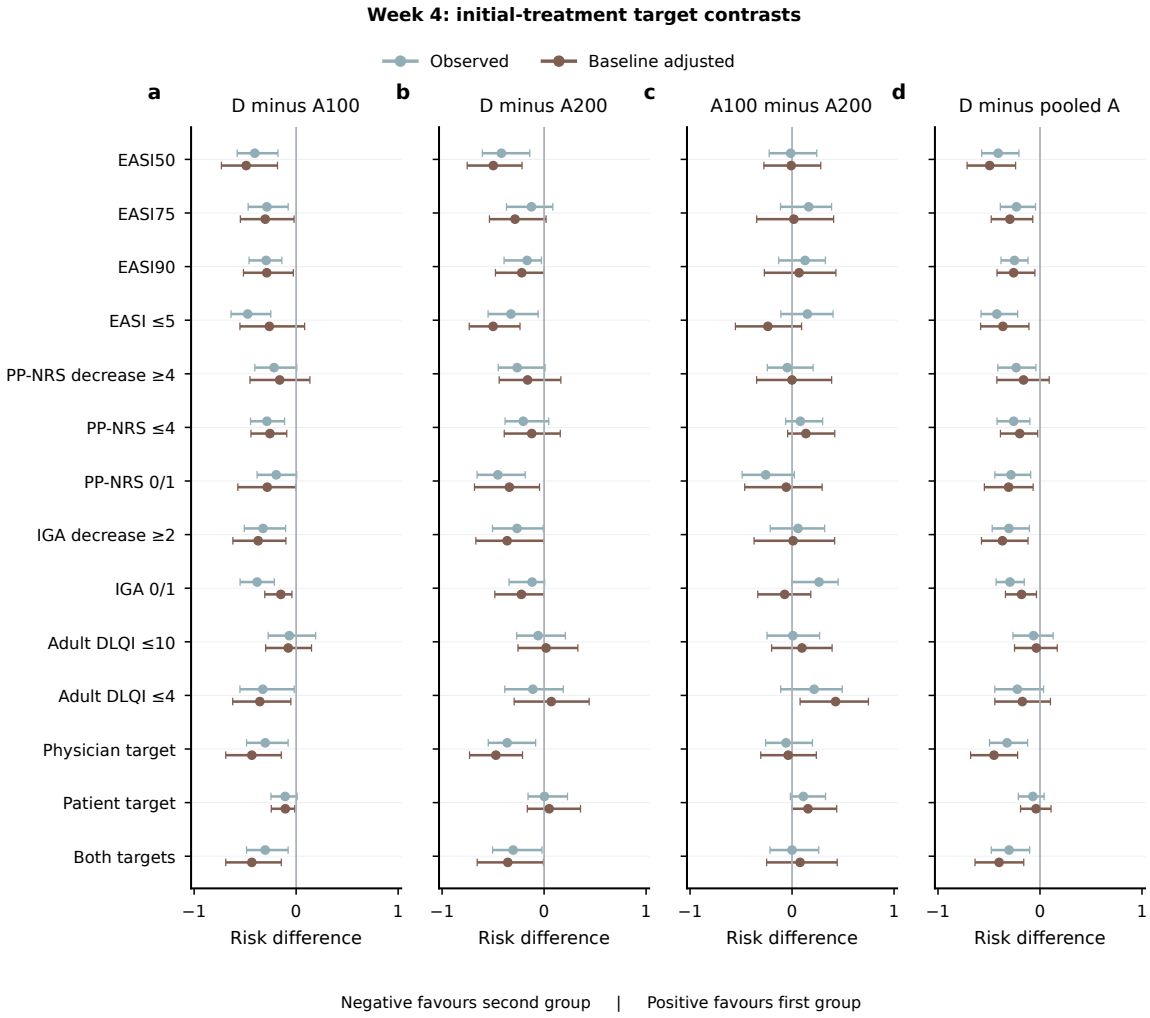


Figure S23: **Week 4: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. D denotes dupilumab; A100/A200 the initial abrocitinib doses. Pooled abrocitinib is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

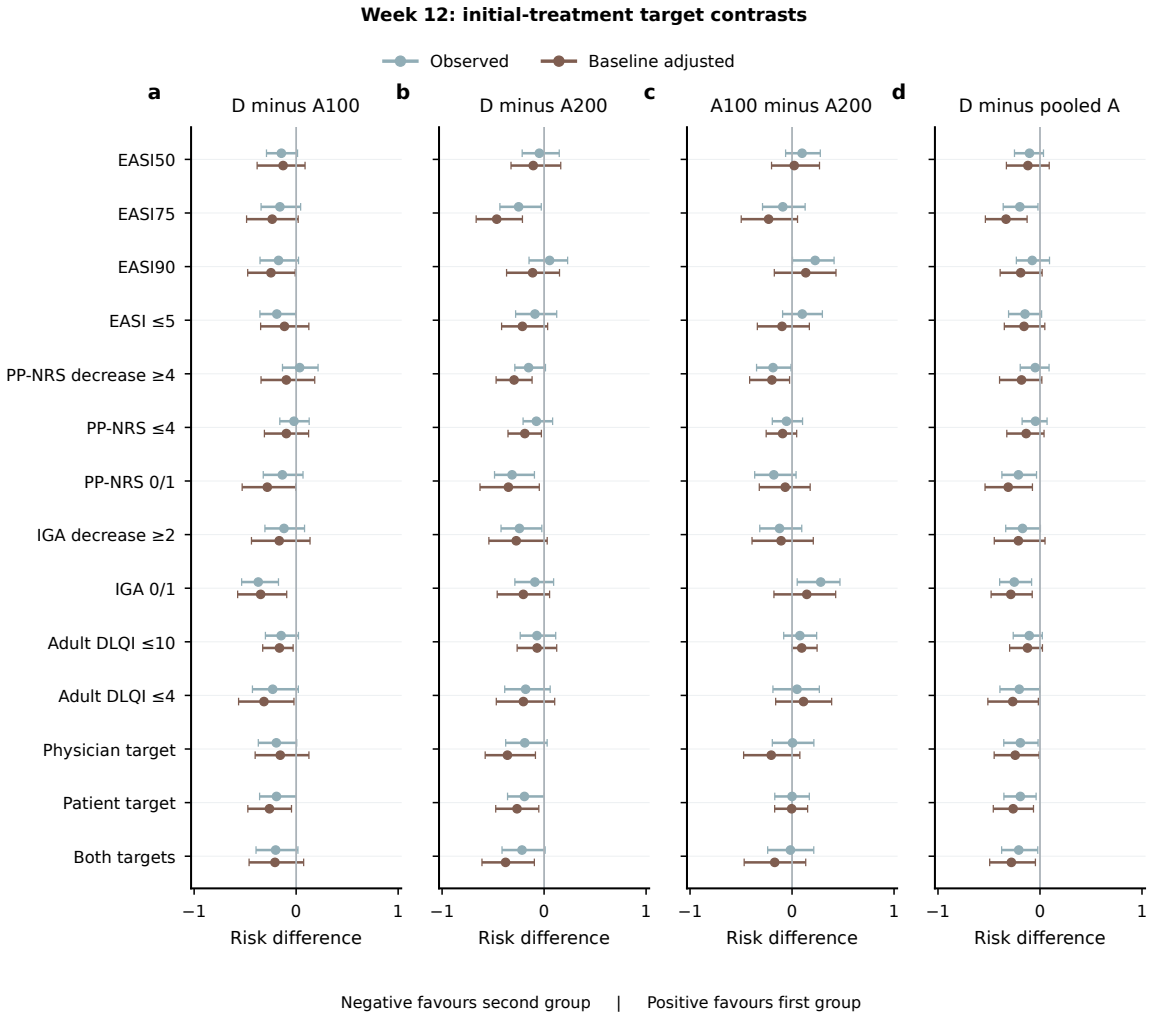


Figure S24: **Week 12: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. D denotes dupilumab; A100/A200 the initial abrocitinib doses. Pooled abrocitinib is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

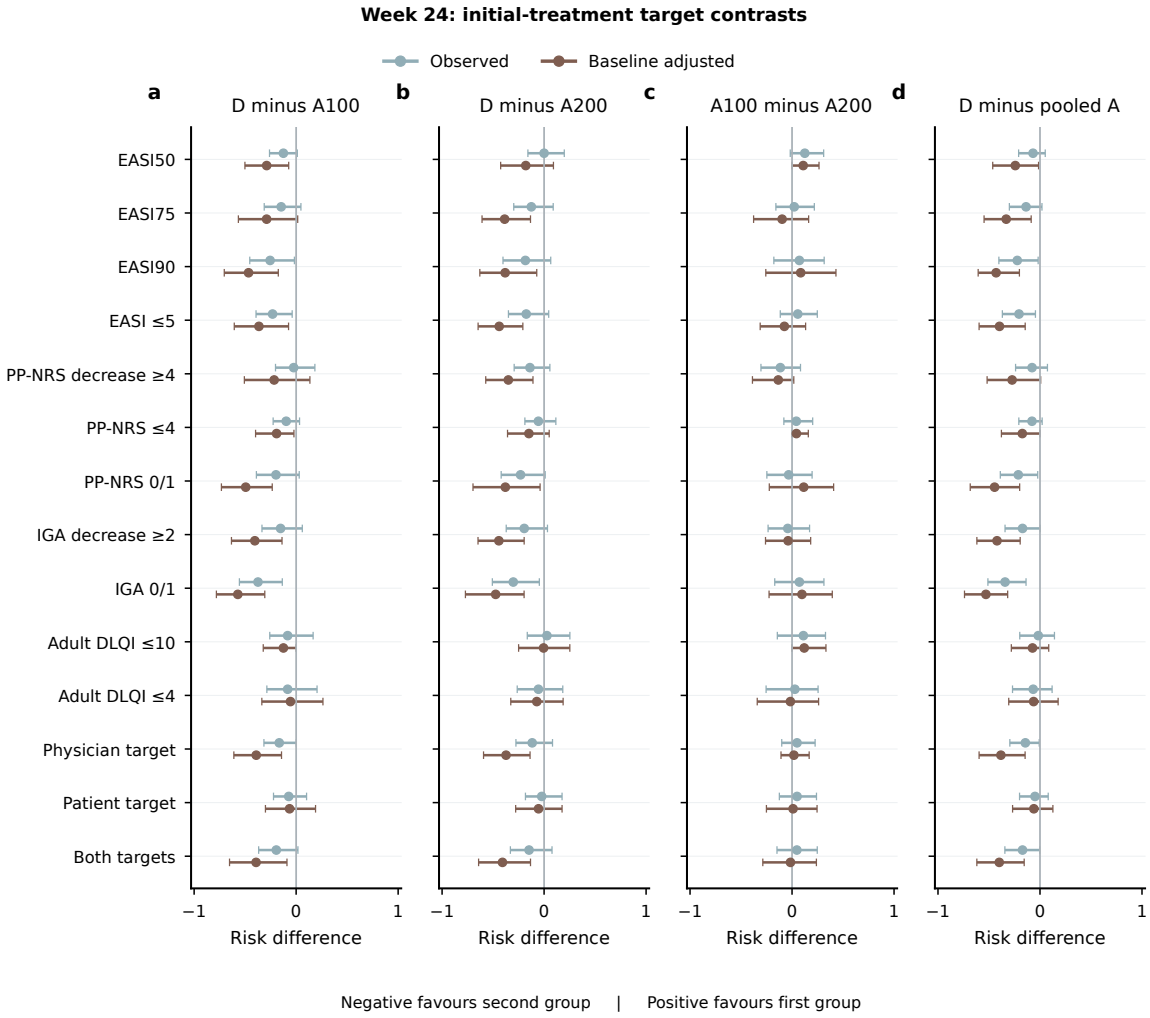


Figure S25: **Week 24: all initial-treatment target contrasts.** Observed and baseline-adjusted risk differences with nominal 95% confidence intervals. Positive values favour the first named treatment for target attainment. Adjustment uses generalized overlap weights from baseline EASI, age, sex, BMI and centre; it does not estimate a randomized current-dose effect. D denotes dupilumab; A100/A200 the initial abrocitinib doses. Pooled abrocitinib is supplementary. Missing estimates are left absent; all exact denominators, Fisher/weighted tests, Holm-adjusted P values and exploratory FDR values are in the source tables.

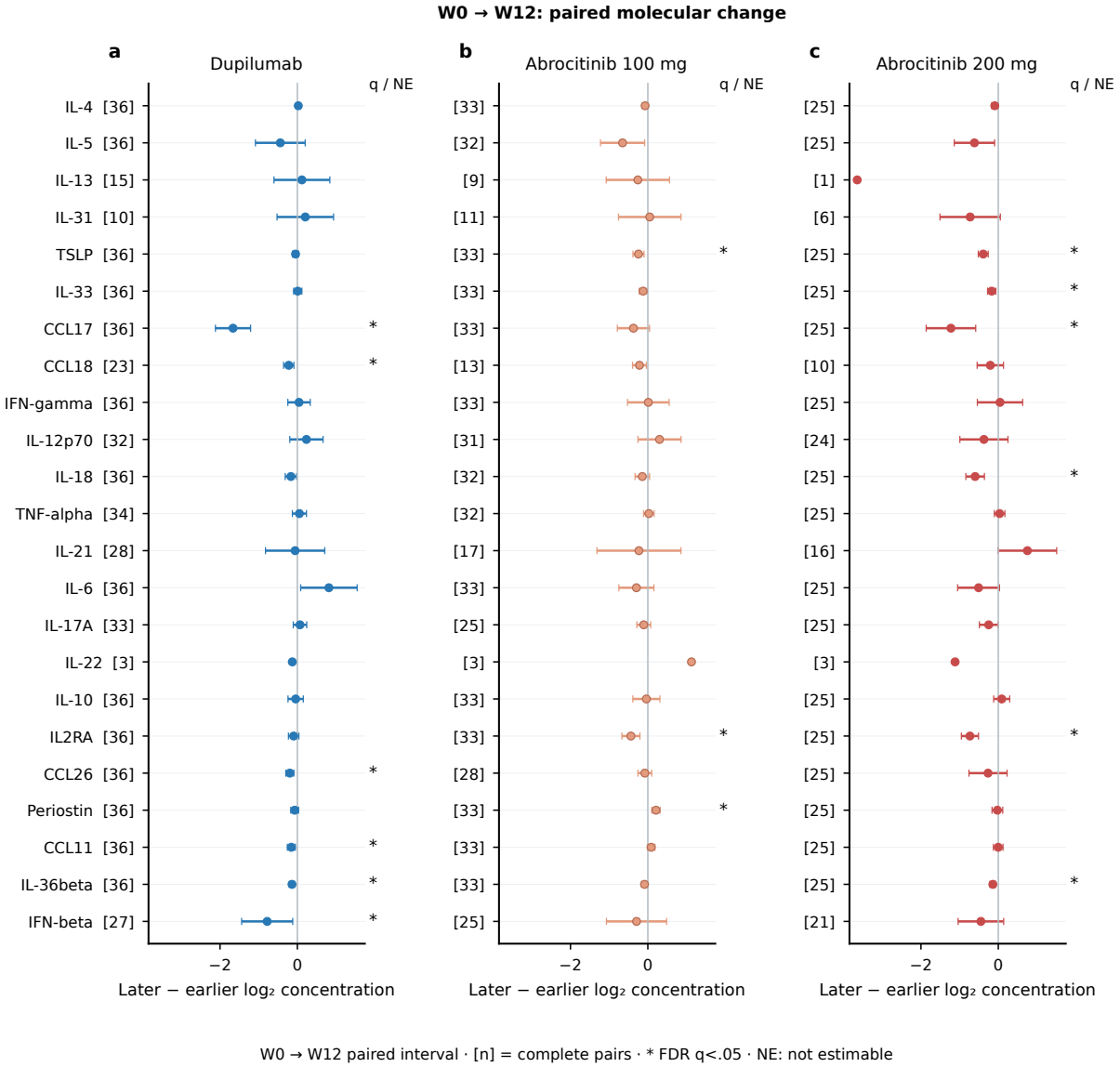


Figure S5: **All molecular changes within W0 → W12.** Mean within-patient $\log_2$ concentration changes and 95% t confidence intervals. Negative values mean a decline: −1 equals a geometric paired ratio of 0.5. Each marker/treatment uses its available pairs, shown as [n]; sets may differ between windows. Points with fewer than five pairs are descriptive and have no inferential interval; rows without an estimable mean are marked NE (not estimable) in the outside status column. Stars indicate BH q<0.05 across the full 207 marker-by-treatment-by-window family. All 23 measured markers are included.

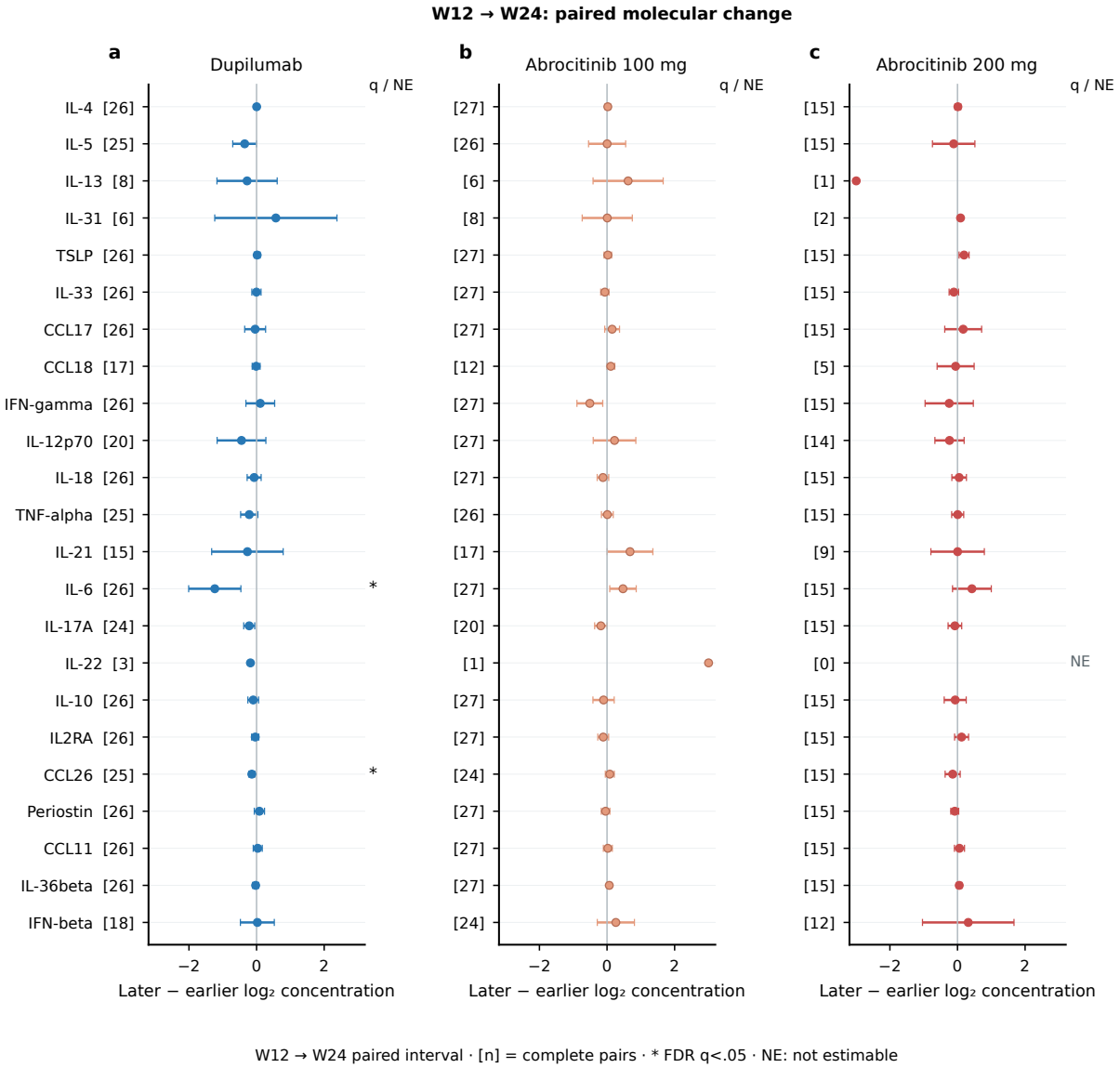


Figure S6: **All molecular changes within W12 → W24.** Mean within-patient $\log_2$ concentration changes and 95% t confidence intervals. Negative values mean a decline: -1 equals a geometric paired ratio of 0.5. Each marker/treatment uses its available pairs, shown as [n]; sets may differ between windows. Points with fewer than five pairs are descriptive and have no inferential interval; rows without an estimable mean are marked NE (not estimable) in the outside status column. Stars indicate BH q<0.05 across the full 207 marker-by-treatment-by-window family. All 23 measured markers are included.

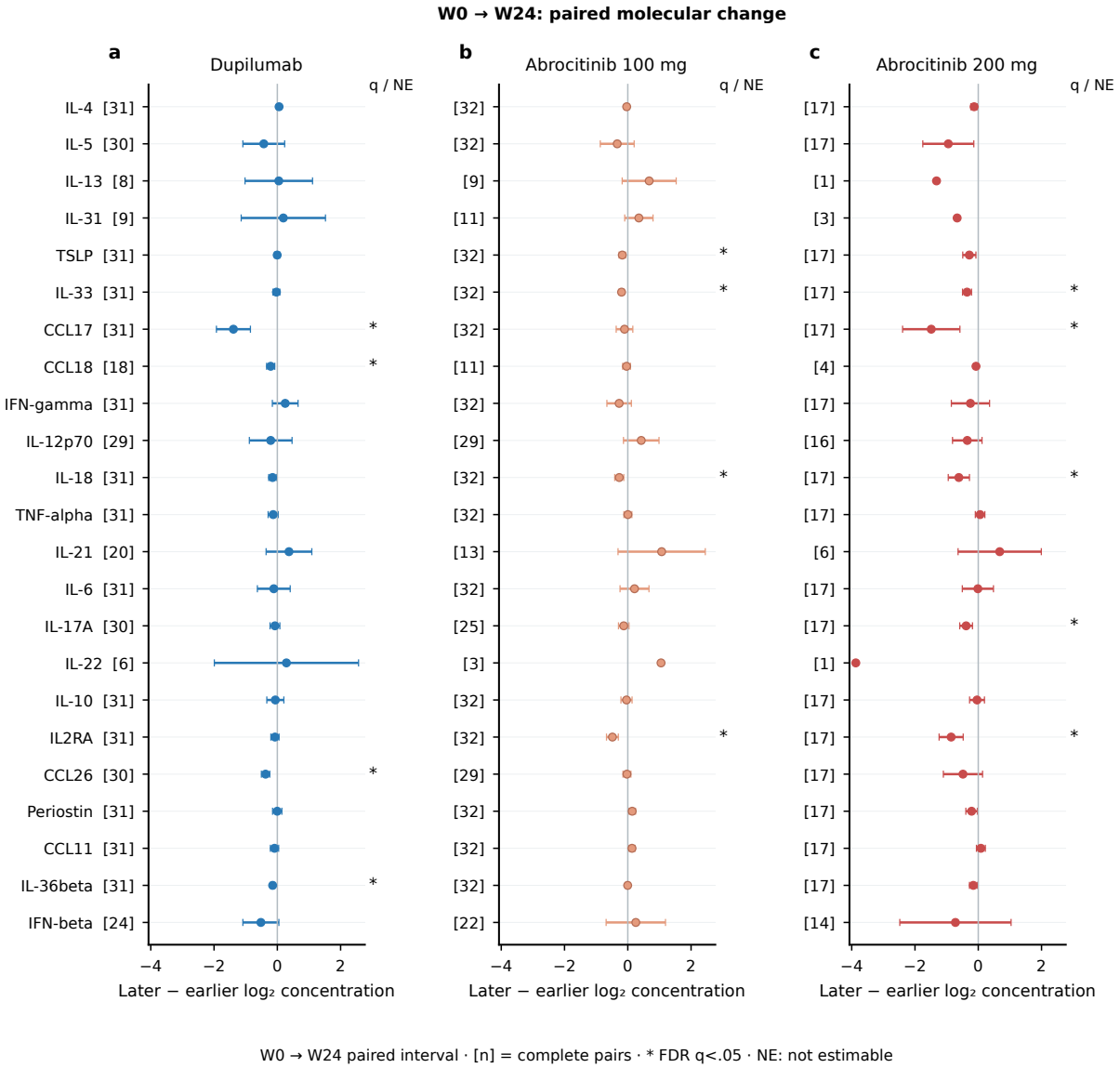


Figure S7: **All molecular changes within W0 → W24.** Mean within-patient $\log_2$ concentration changes and 95% t confidence intervals. Negative values mean a decline: -1 equals a geometric paired ratio of 0.5. Each marker/treatment uses its available pairs, shown as [n]; sets may differ between windows. Points with fewer than five pairs are descriptive and have no inferential interval; rows without an estimable mean are marked NE (not estimable) in the outside status column. Stars indicate BH q<0.05 across the full 207 marker-by-treatment-by-window family. All 23 measured markers are included.

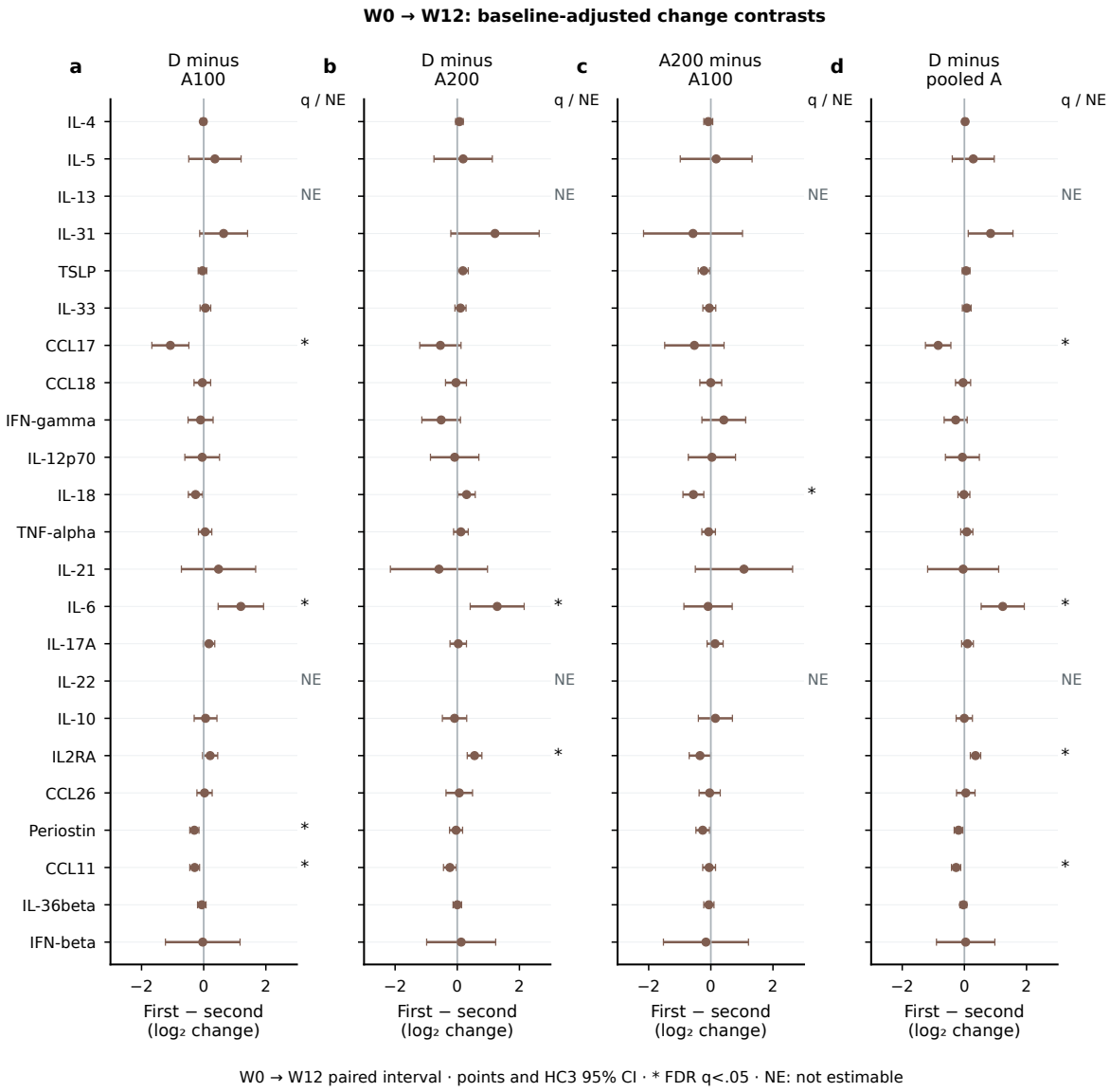


Figure S8: **Adjusted treatment contrasts in molecular change: W0 → W12.** Baseline-adjusted contrasts with HC3 95% confidence intervals. All 23 markers and four contrasts are shown. D denotes dupilumab and A100/A200 initial abrocitinib doses. Negative means the first treatment has a smaller, more negative change; it is not a negative concentration. Models control starting $\log_2$ concentration, baseline total EASI, age, sex, BMI and centre; source tables provide exact n, exclusions, BH q values, centre and batch sensitivities. Pooled abrocitinib retains the observed initial-dose mixture. These are observational initial-treatment contrasts, not randomized current-dose effects.

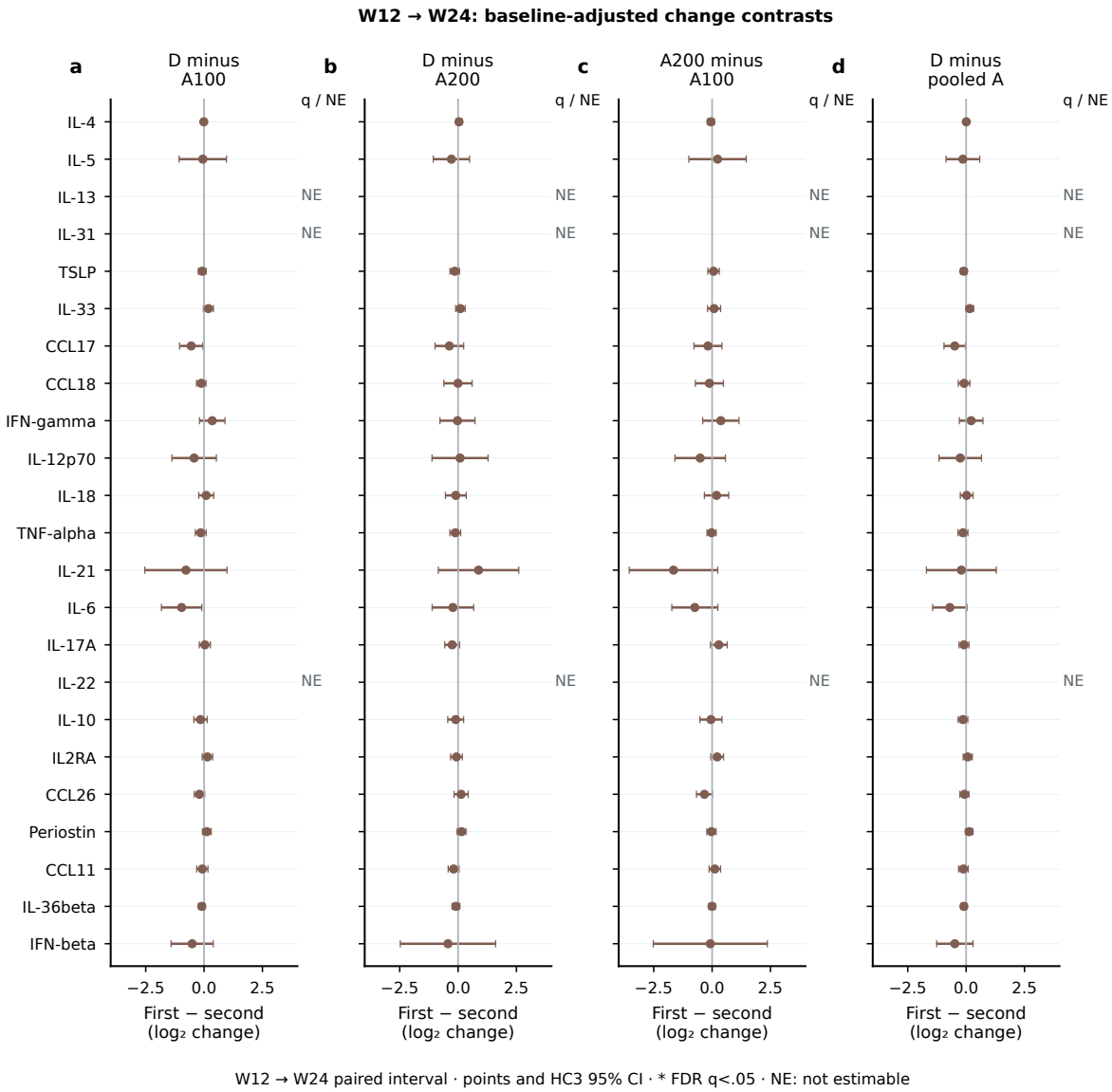


Figure S9: **Adjusted treatment contrasts in molecular change: W12 → W24.** Baseline-adjusted contrasts with HC3 95% confidence intervals. All 23 markers and four contrasts are shown. D denotes dupilumab and A100/A200 initial abrocitinib doses. Negative means the first treatment has a smaller, more negative change; it is not a negative concentration. Models control starting $\log_2$ concentration, baseline total EASI, age, sex, BMI and centre; source tables provide exact n, exclusions, BH q values, centre and batch sensitivities. Pooled abrocitinib retains the observed initial-dose mixture. These are observational initial-treatment contrasts, not randomized current-dose effects.

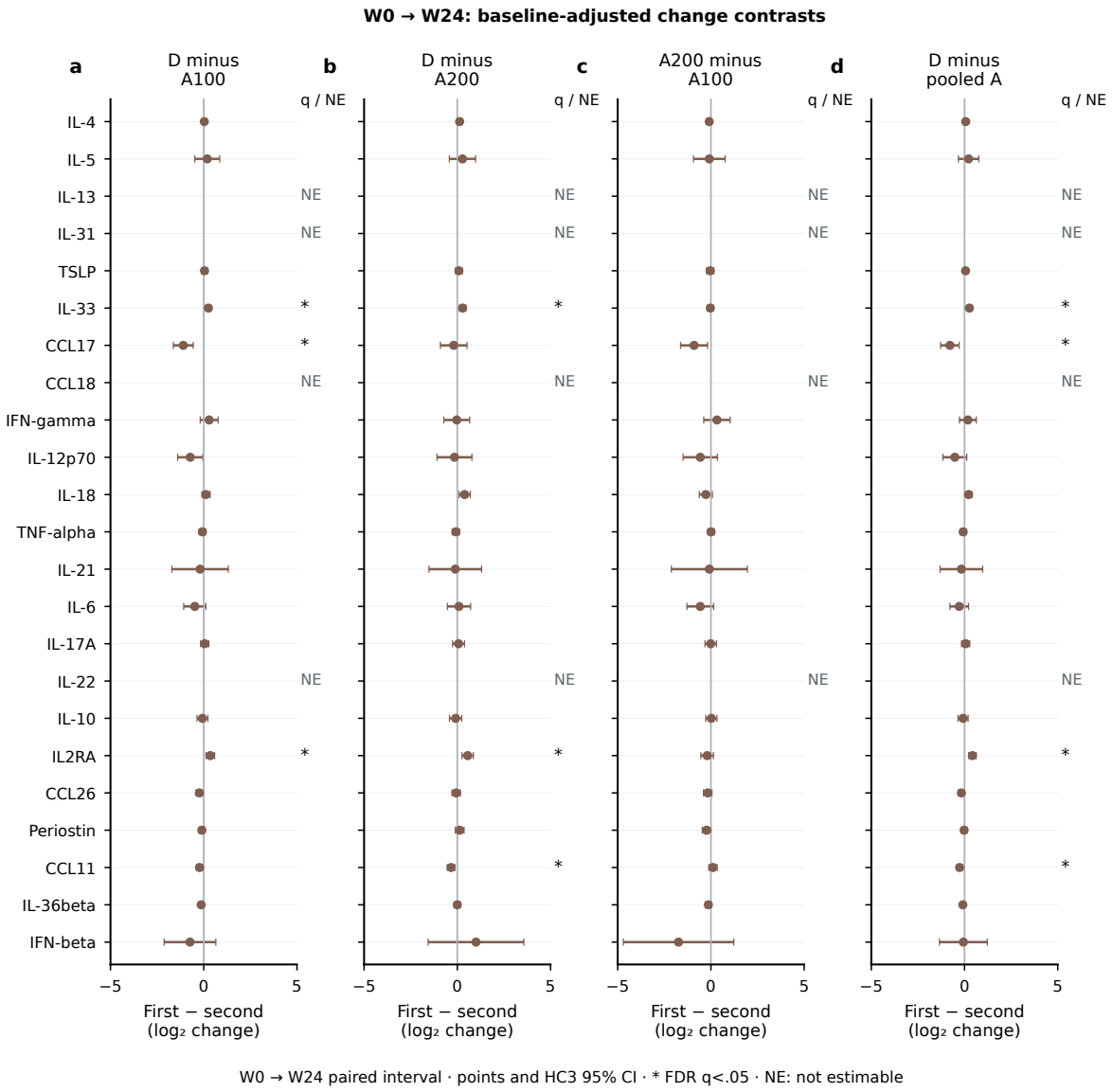


Figure S10: **Adjusted treatment contrasts in molecular change: W0 → W24.** Baseline-adjusted contrasts with HC3 95% confidence intervals. All 23 markers and four contrasts are shown. D denotes dupilumab and A100/A200 initial abrocitinib doses. Negative means the first treatment has a smaller, more negative change; it is not a negative concentration. Models control starting $\log_2$ concentration, baseline total EASI, age, sex, BMI and centre; source tables provide exact n, exclusions, BH q values, centre and batch sensitivities. Pooled abrocitinib retains the observed initial-dose mixture. These are observational initial-treatment contrasts, not randomized current-dose effects.

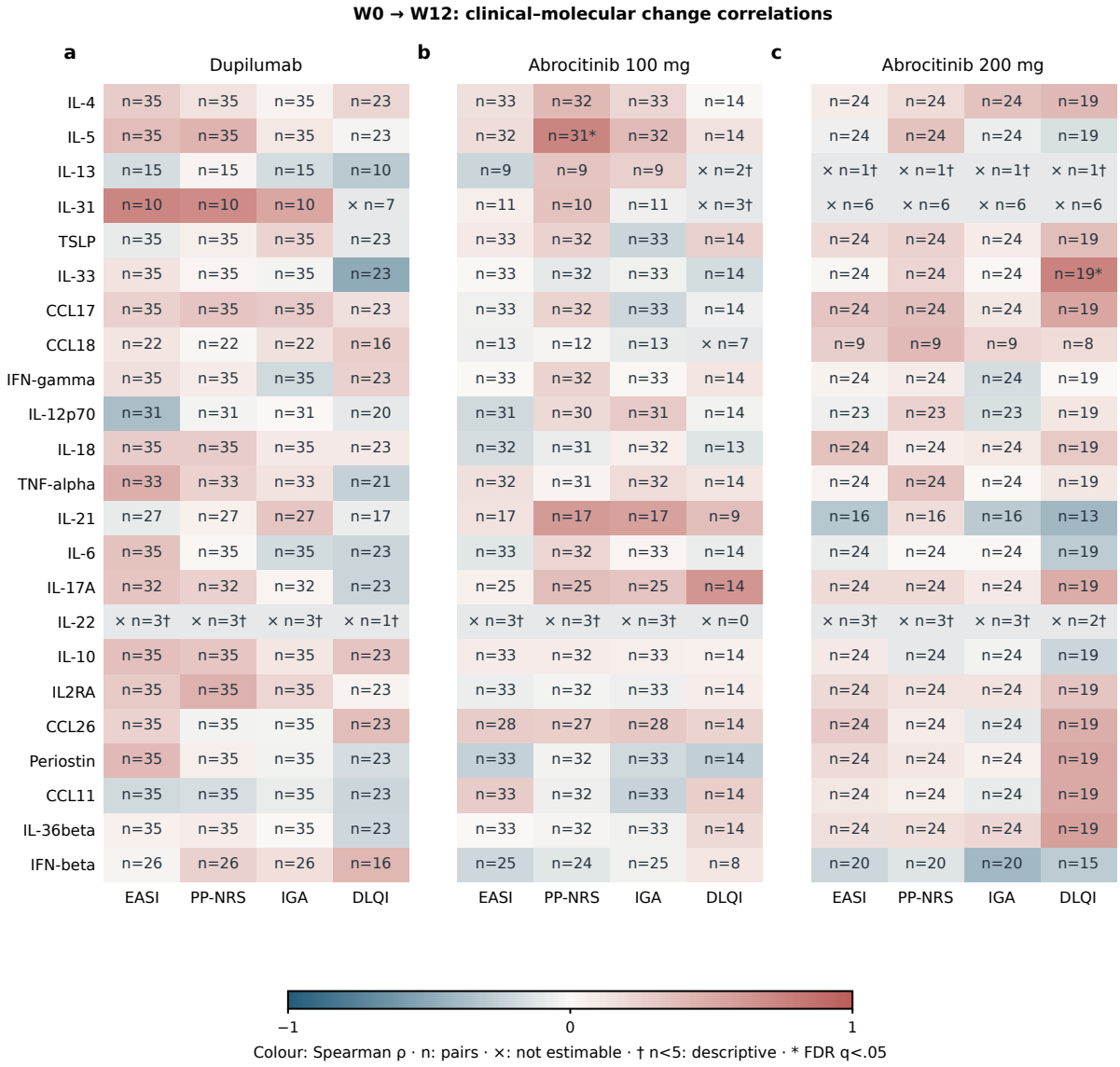


Figure S11: **Clinical–molecular change associations: W0 → W12.** Colour is Spearman correlation between paired clinical change and paired $\log_2$ molecular change, both later minus earlier. Cell numbers show complete-pair patient counts and stars BH q<0.05. Positive correlations indicate changes in the same direction; negative correlations indicate opposite changes. Cells marked × have no estimable correlation; printed n distinguishes no pairs from an unestimable coefficient among available pairs. Daggers flag descriptive small cells with n<5. Adult DLQI requires age ≥18. Correlation does not establish numerical interchangeability or causal effects. All coefficients, bootstrap confidence intervals and multiplicity-adjusted tests remain in source data.

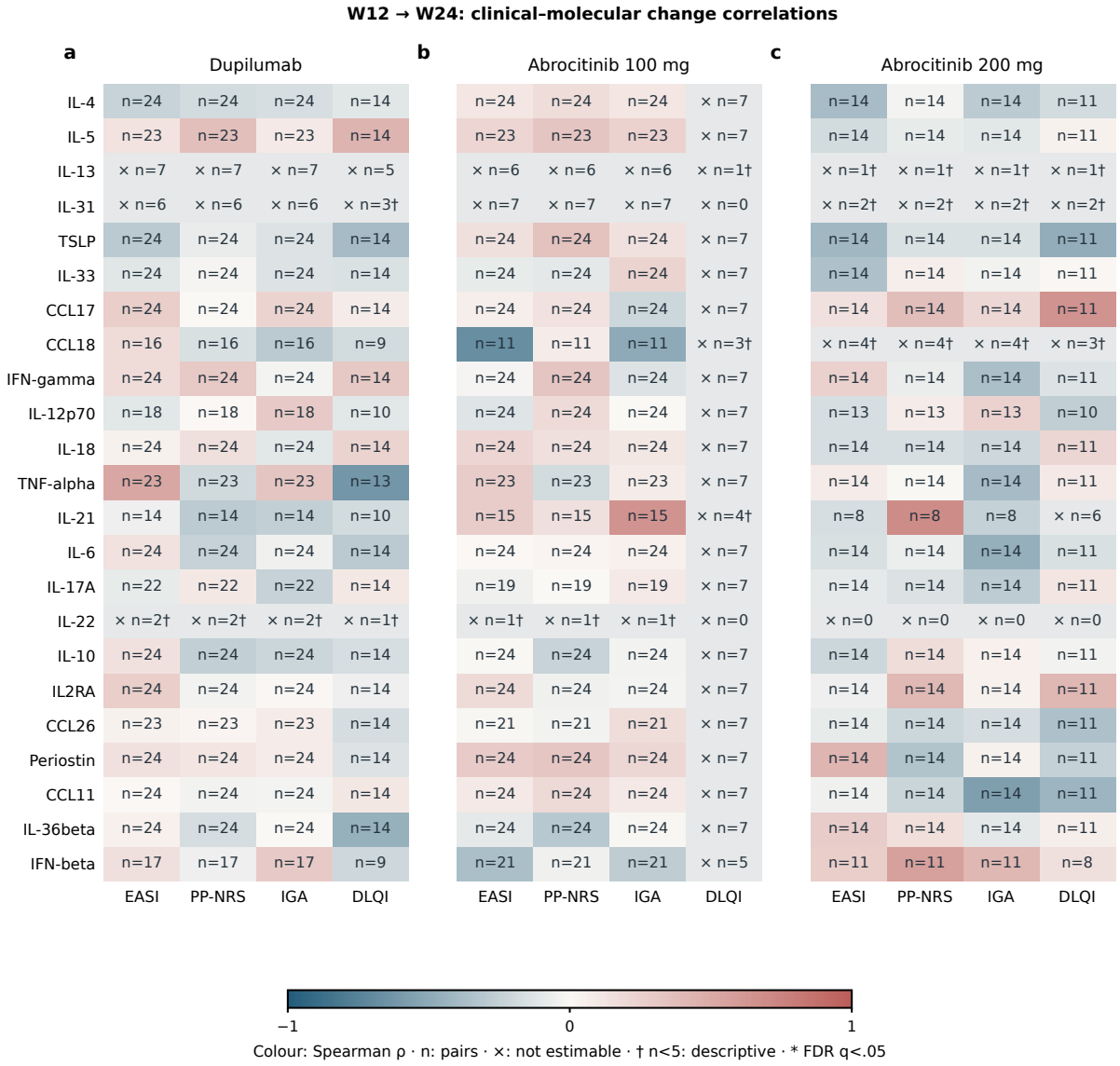


Figure S12: **Clinical–molecular change associations: W12 → W24.** Colour is Spearman correlation between paired clinical change and paired $\log_2$ molecular change, both later minus earlier. Cell numbers show complete-pair patient counts and stars BH q<0.05. Positive correlations indicate changes in the same direction; negative correlations indicate opposite changes. Cells marked × have no estimable correlation; printed n distinguishes no pairs from an unestimable coefficient among available pairs. Daggers flag descriptive small cells with n<5. Adult DLQI requires age ≥18. Correlation does not establish numerical interchangeability or causal effects. All coefficients, bootstrap confidence intervals and multiplicity-adjusted tests remain in source data.

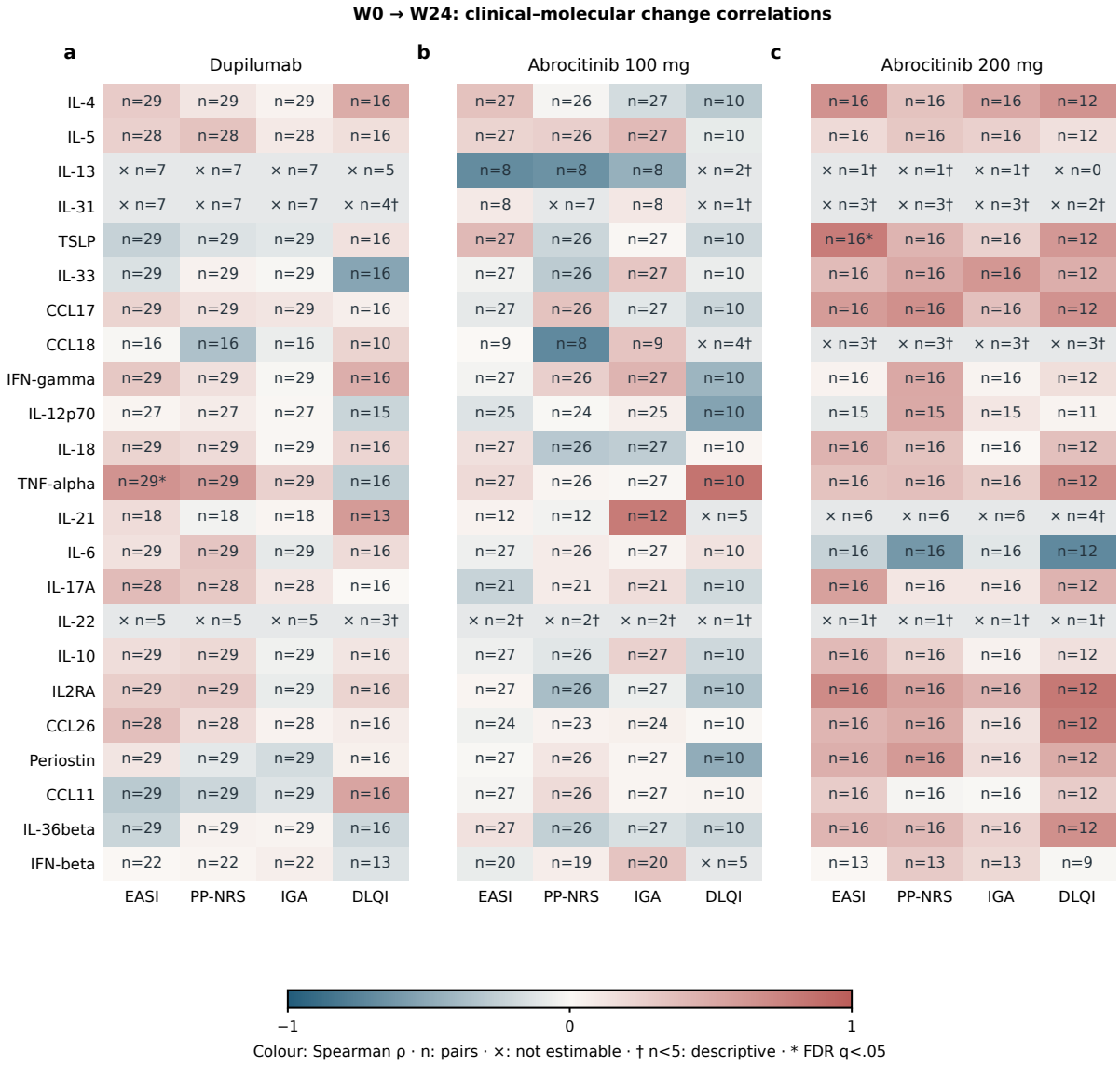


Figure S13: **Clinical–molecular change associations: W0 → W24.** Colour is Spearman correlation between paired clinical change and paired $\log_2$ molecular change, both later minus earlier. Cell numbers show complete-pair patient counts and stars BH q<0.05. Positive correlations indicate changes in the same direction; negative correlations indicate opposite changes. Cells marked × have no estimable correlation; printed n distinguishes no pairs from an unestimable coefficient among available pairs. Daggers flag descriptive small cells with n<5. Adult DLQI requires age ≥18. Correlation does not establish numerical interchangeability or causal effects. All coefficients, bootstrap confidence intervals and multiplicity-adjusted tests remain in source data.

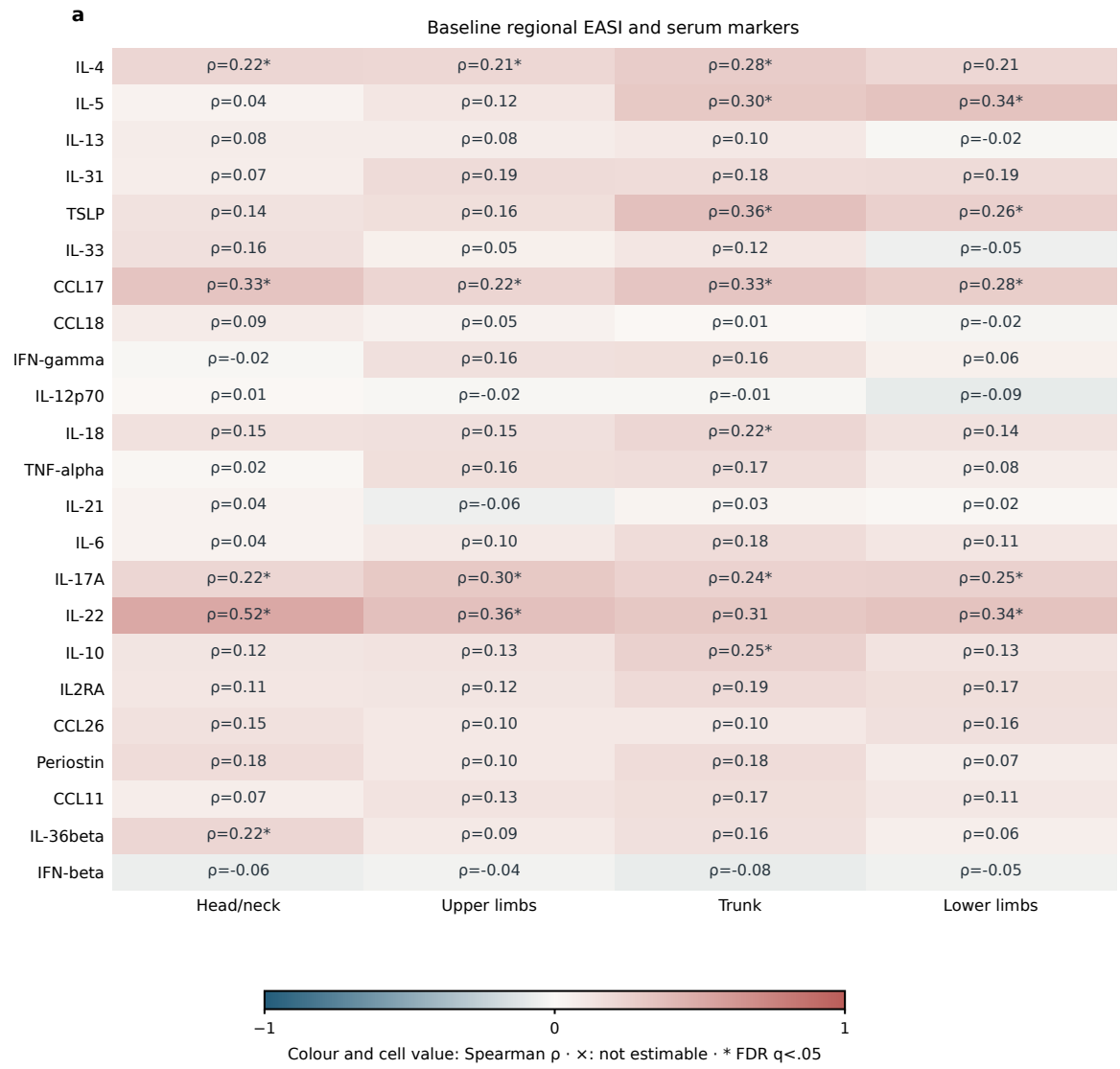


Figure S14: **Baseline molecular associations with four skin regions.** Observed baseline Spearman correlations pooled across initial-treatment groups, using source-consistent regional EASI and matched serum measurements. Values are printed in cells; stars indicate BH q<0.05. Regional records must be complete, within theoretical bounds and sum to total EASI within 0.01. Counts and bootstrap confidence intervals are in the source table. Serum association alone does not establish site-specific pathogenesis.

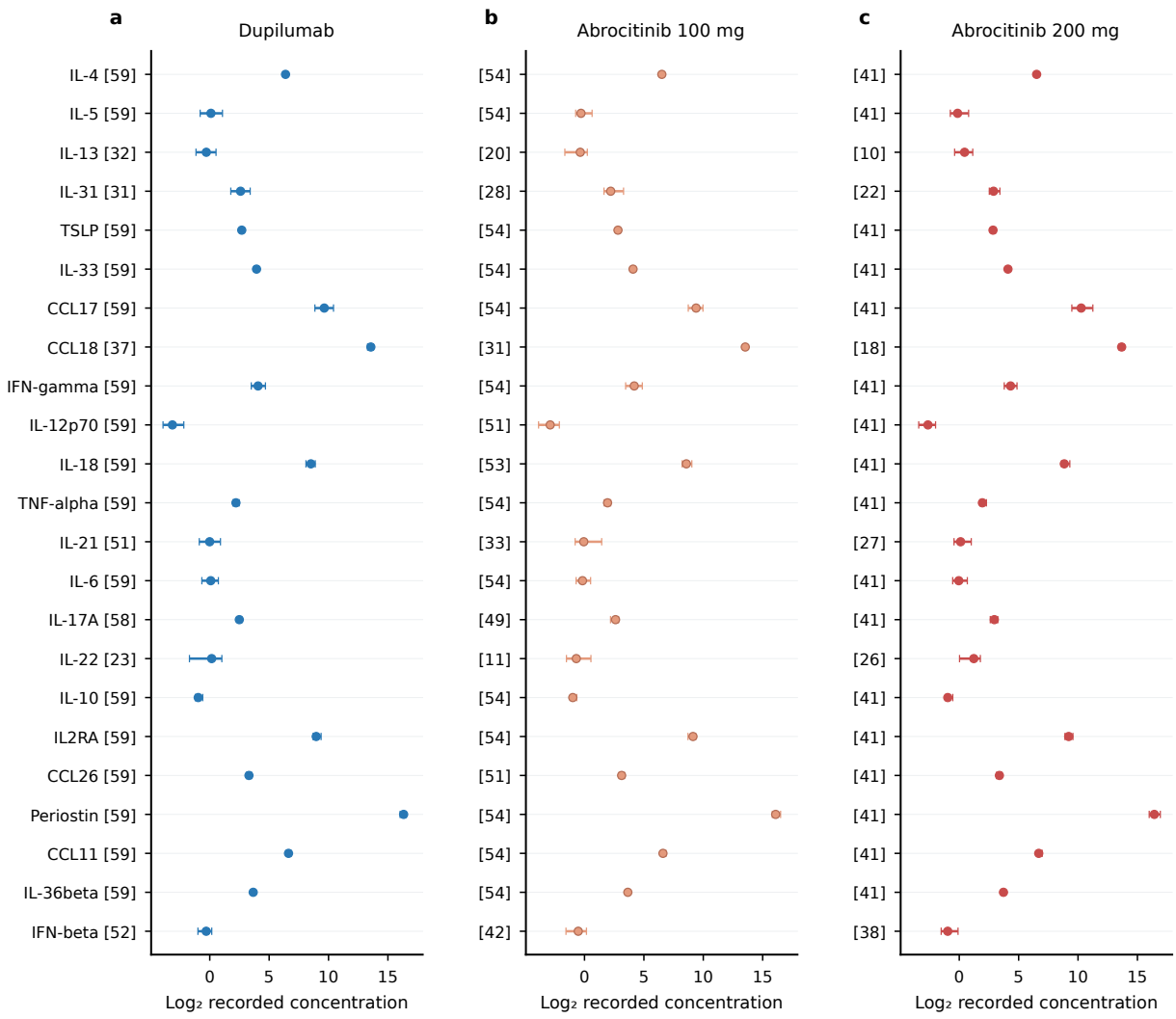


Figure S18: **Baseline concentration distributions for the full measured panel.** Baseline medians and interquartile ranges of $\log_2$ recorded concentrations, with available patient counts [n]. Treatment colours are fixed across the manuscript. Compare treatment groups within each marker; absolute values across different analytes do not represent comparable biological activity. All 23 markers are retained, including sparse or censored assays. Censoring and repeated values are separately audited.

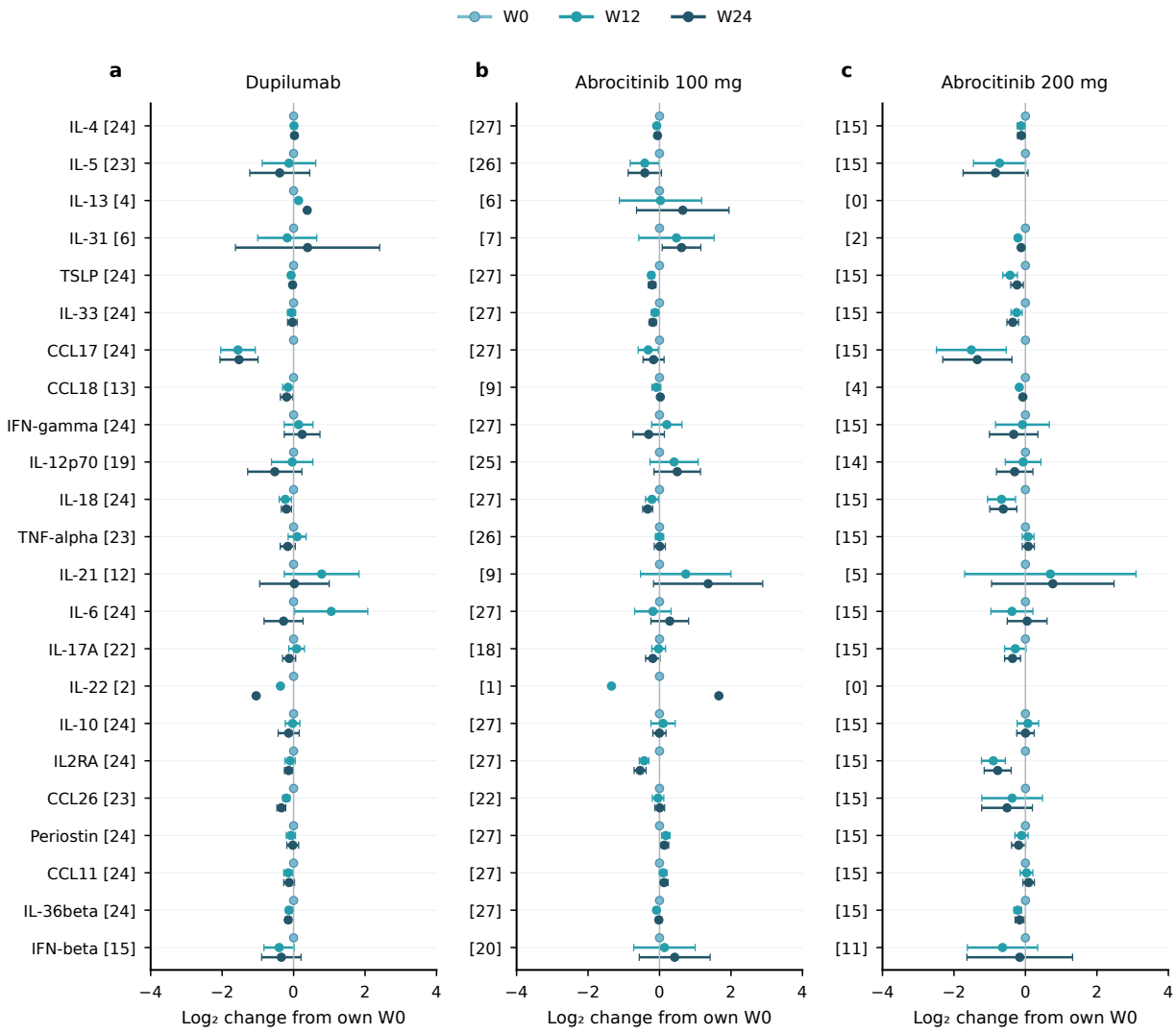


Figure S19: **Molecular W0/W12/W24 changes in identical patient sets.** Within each marker and initial treatment, all three colours use the same patients observed at W0/W12/W24. Fixed triplet counts are shown as [n]. W0 equals zero by definition; W12/W24 are mean within-patient $\log_2$ changes from W0 with 95% t confidence intervals when n≥5. Sets with n<5 are descriptive and have no inferential interval. Negative values indicate lower concentration, not absence of the analyte. This includes every measured marker and its uncertainty, complementing the overview heatmap.

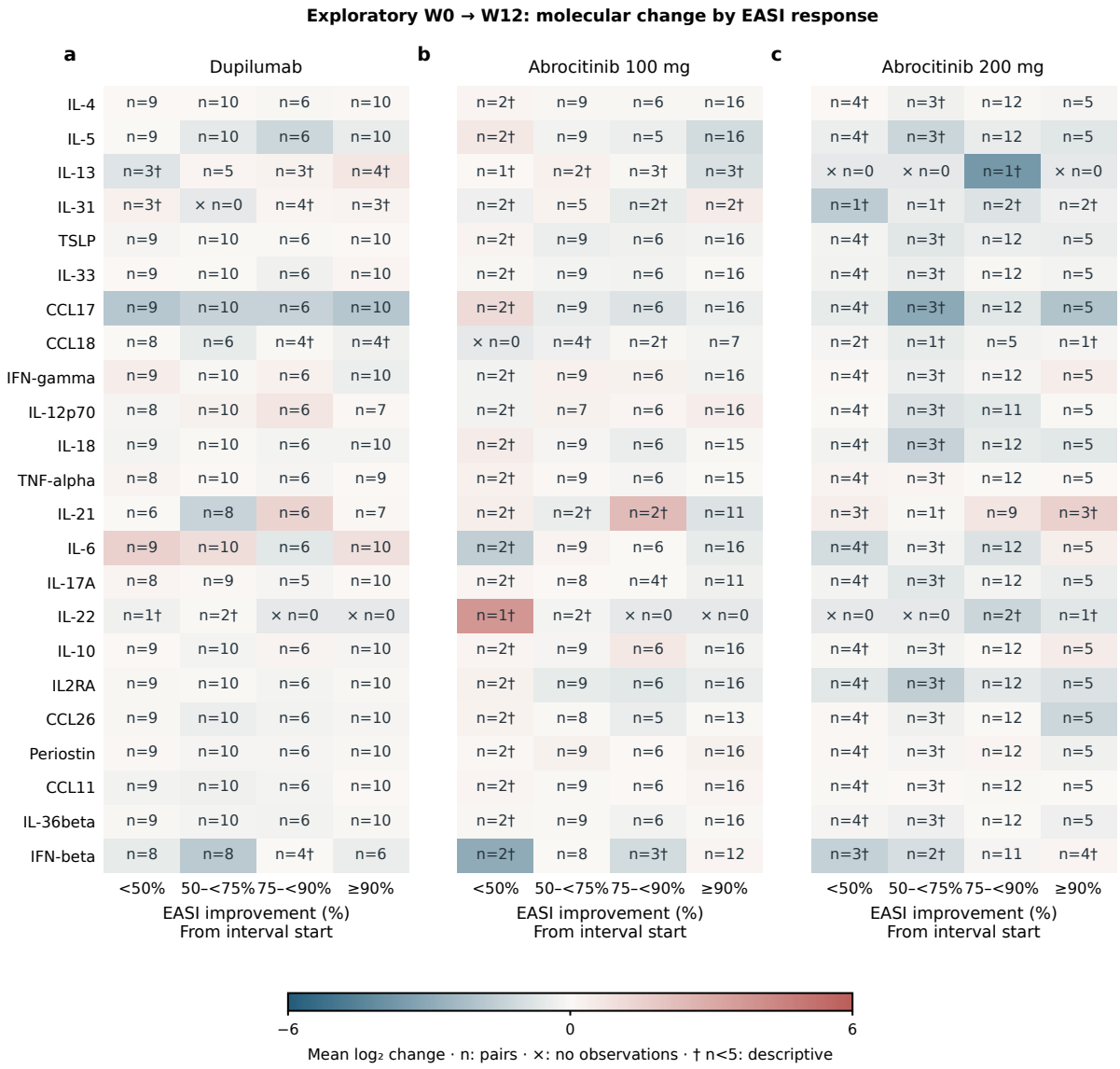


Figure S26: **Molecular changes by EASI response: W0 → W12.** Columns are mutually exclusive clinical-response categories within the named paired interval. Colour is mean within-patient $\log_2$ concentration change; numbers are complete-pair patient counts, with a dagger indicating descriptive small cells with n<5. Negative means lower later concentration. These are neither correlation coefficients nor change per clinical point. EASI improvement is relative to the start of the named interval; IGA uses recorded ordinal decrease. Cells marked × and n=0 have no observations; absence is not a zero molecular change. All 23 markers, exact means and intervals are retained in the source table; the small cells are descriptive and do not validate a prediction model.

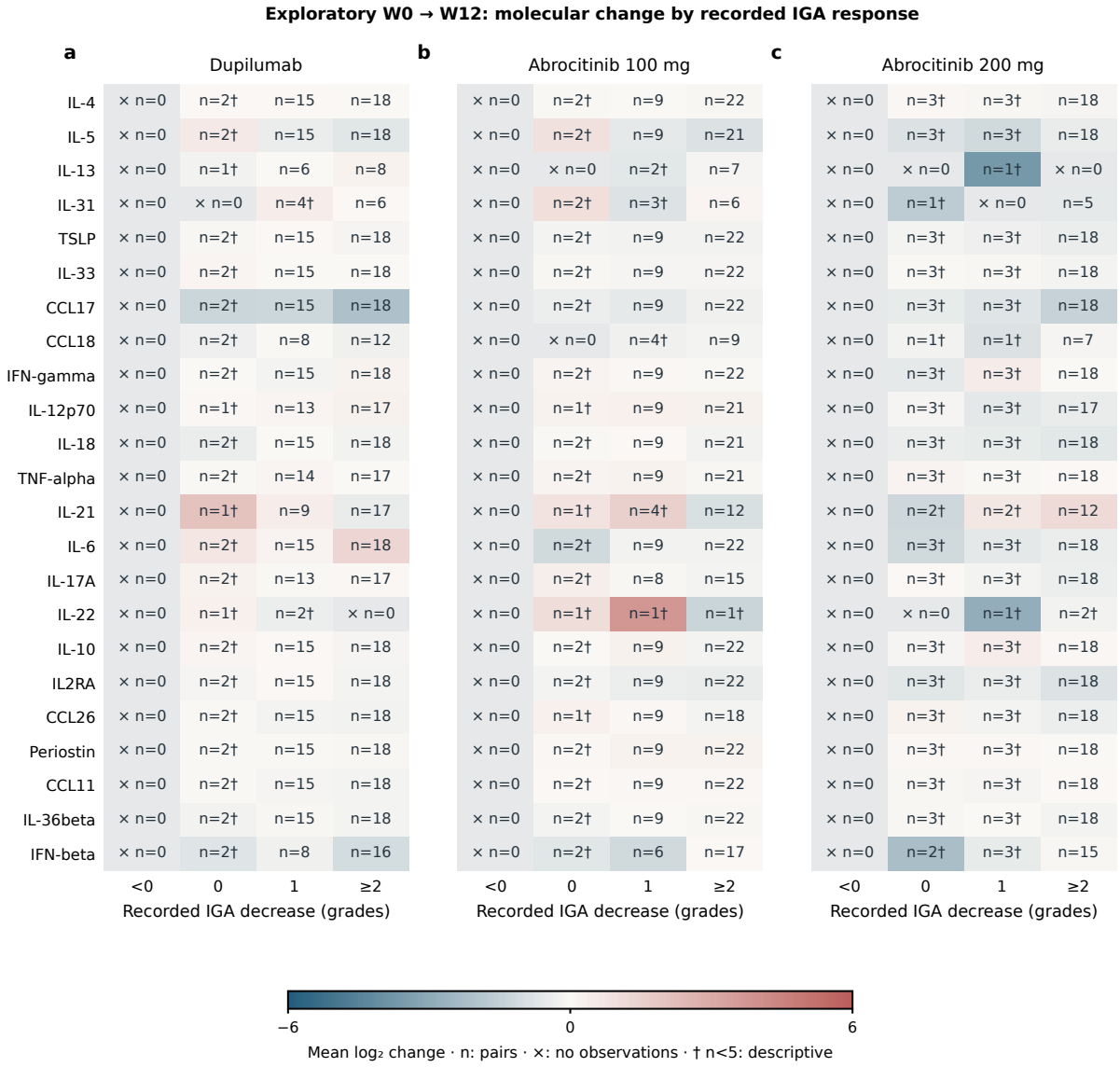


Figure S27: **Molecular changes by IGA response: W0 → W12.** Columns are mutually exclusive clinical-response categories within the named paired interval. Colour is mean within-patient $\log_2$ concentration change; numbers are complete-pair patient counts, with a dagger indicating descriptive small cells with n<5. Negative means lower later concentration. These are neither correlation coefficients nor change per clinical point. EASI improvement is relative to the start of the named interval; IGA uses recorded ordinal decrease. Cells marked × and n=0 have no observations; absence is not a zero molecular change. All 23 markers, exact means and intervals are retained in the source table; the small cells are descriptive and do not validate a prediction model.

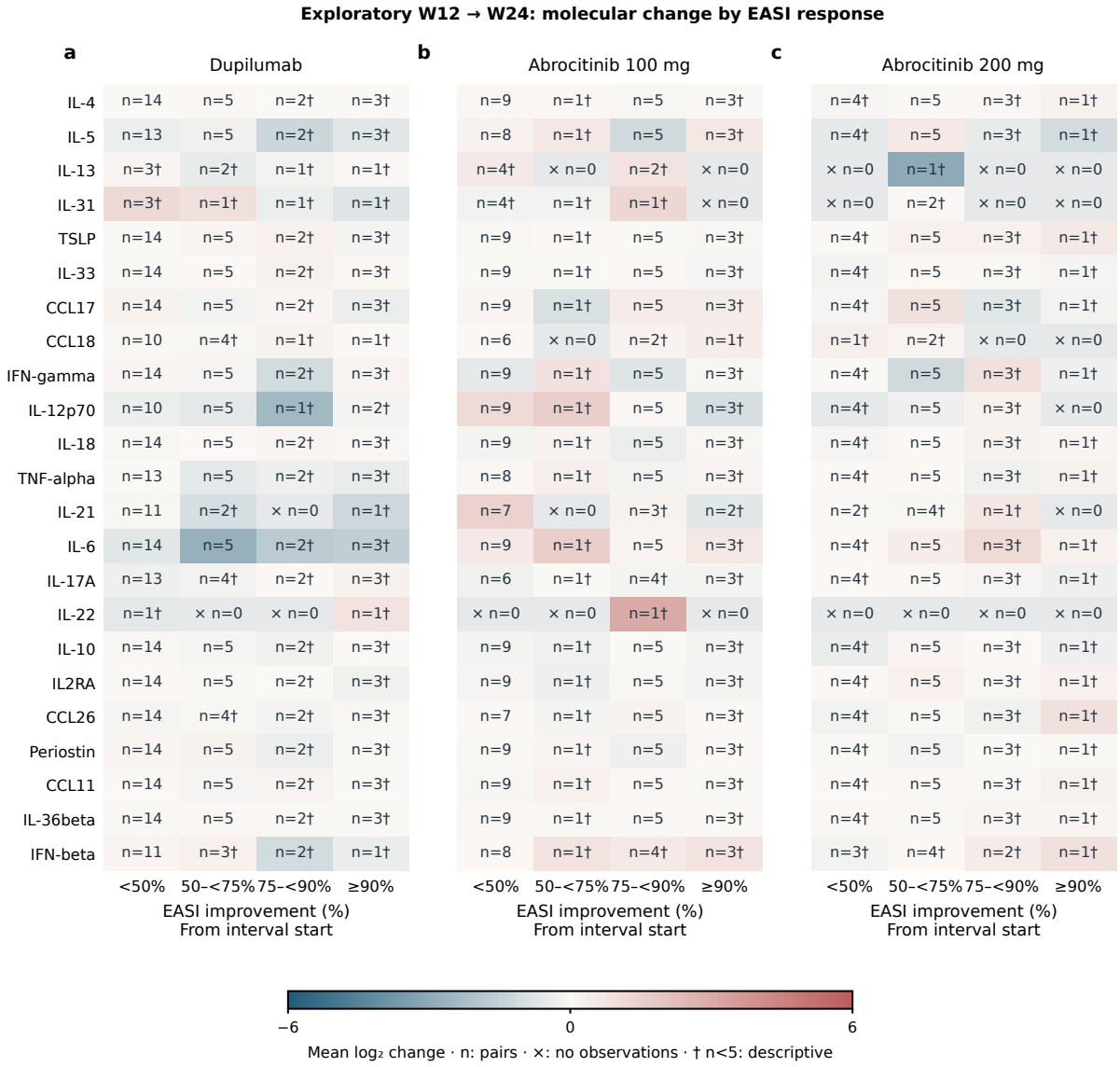


Figure S28: **Molecular changes by EASI response: W12 → W24.** Columns are mutually exclusive clinical-response categories within the named paired interval. Colour is mean within-patient $\log_2$ concentration change; numbers are complete-pair patient counts, with a dagger indicating descriptive small cells with n<5. Negative means lower later concentration. These are neither correlation coefficients nor change per clinical point. EASI improvement is relative to the start of the named interval; IGA uses recorded ordinal decrease. Cells marked × and n=0 have no observations; absence is not a zero molecular change. All 23 markers, exact means and intervals are retained in the source table; the small cells are descriptive and do not validate a prediction model.

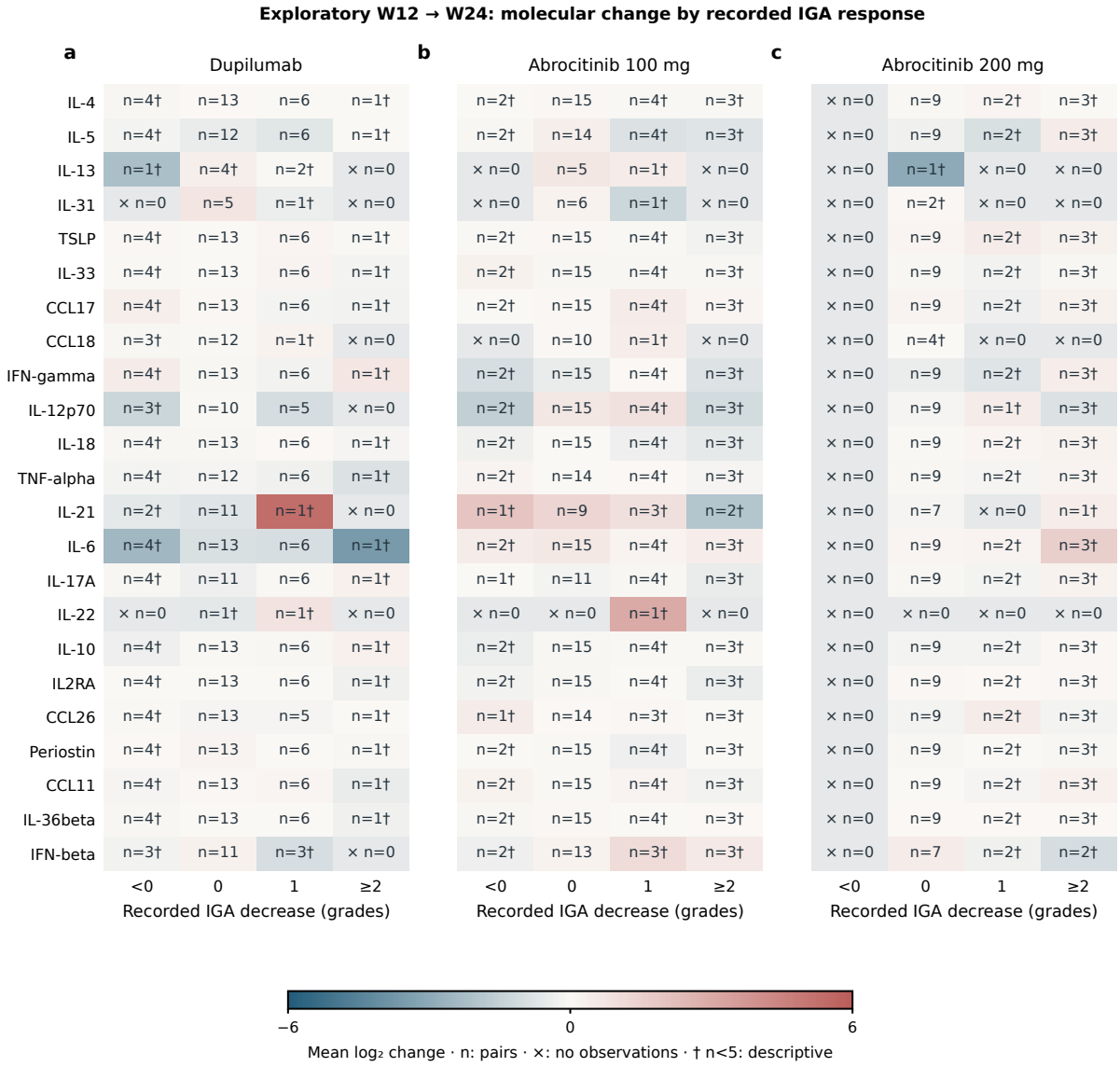


Figure S29: **Molecular changes by IGA response: W12 → W24.** Columns are mutually exclusive clinical-response categories within the named paired interval. Colour is mean within-patient $\log_2$ concentration change; numbers are complete-pair patient counts, with a dagger indicating descriptive small cells with n<5. Negative means lower later concentration. These are neither correlation coefficients nor change per clinical point. EASI improvement is relative to the start of the named interval; IGA uses recorded ordinal decrease. Cells marked × and n=0 have no observations; absence is not a zero molecular change. All 23 markers, exact means and intervals are retained in the source table; the small cells are descriptive and do not validate a prediction model.

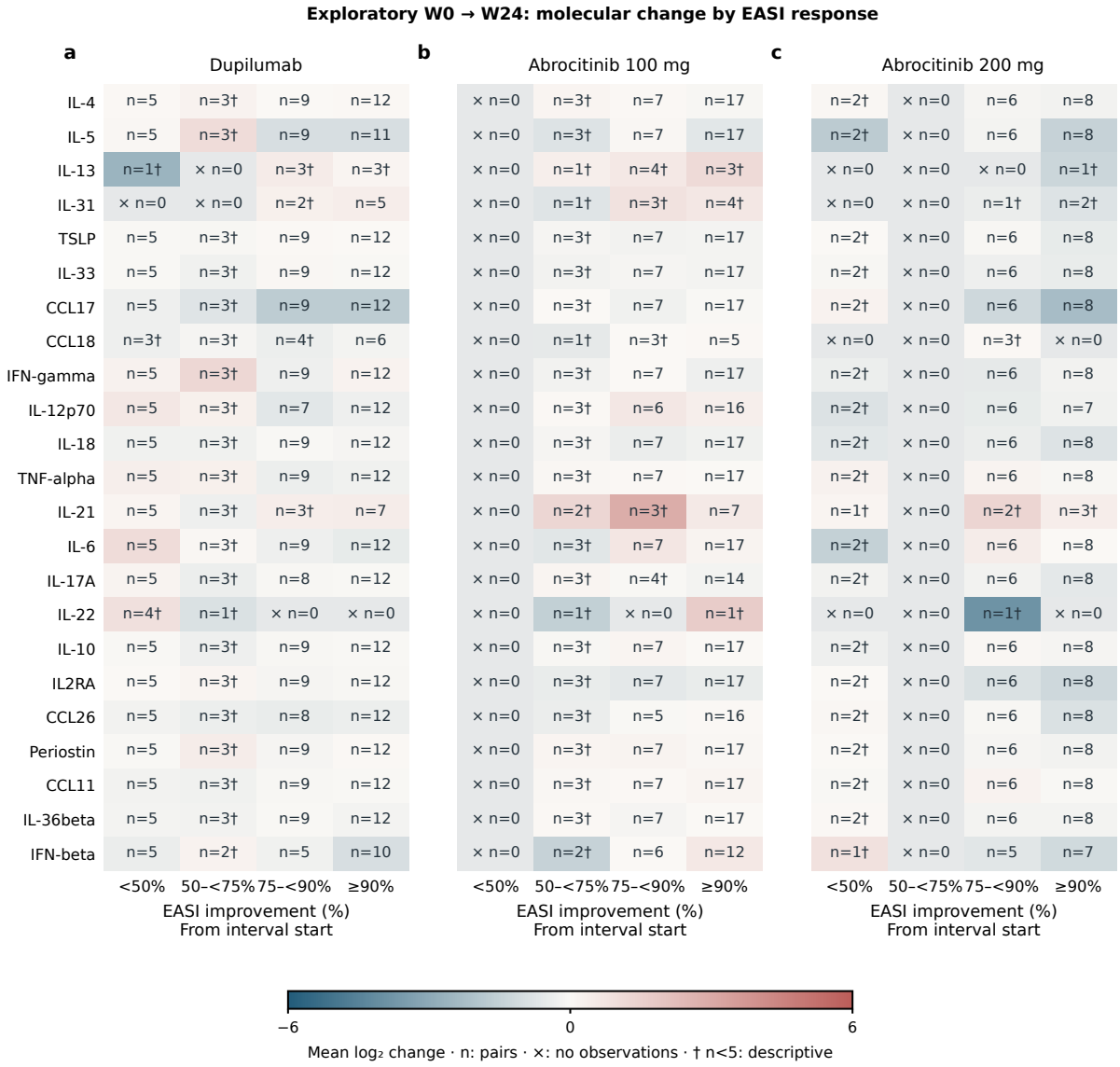


Figure S30: **Molecular changes by EASI response: W0 → W24.** Columns are mutually exclusive clinical-response categories within the named paired interval. Colour is mean within-patient $\log_2$ concentration change; numbers are complete-pair patient counts, with a dagger indicating descriptive small cells with n<5. Negative means lower later concentration. These are neither correlation coefficients nor change per clinical point. EASI improvement is relative to the start of the named interval; IGA uses recorded ordinal decrease. Cells marked × and n=0 have no observations; absence is not a zero molecular change. All 23 markers, exact means and intervals are retained in the source table; the small cells are descriptive and do not validate a prediction model.

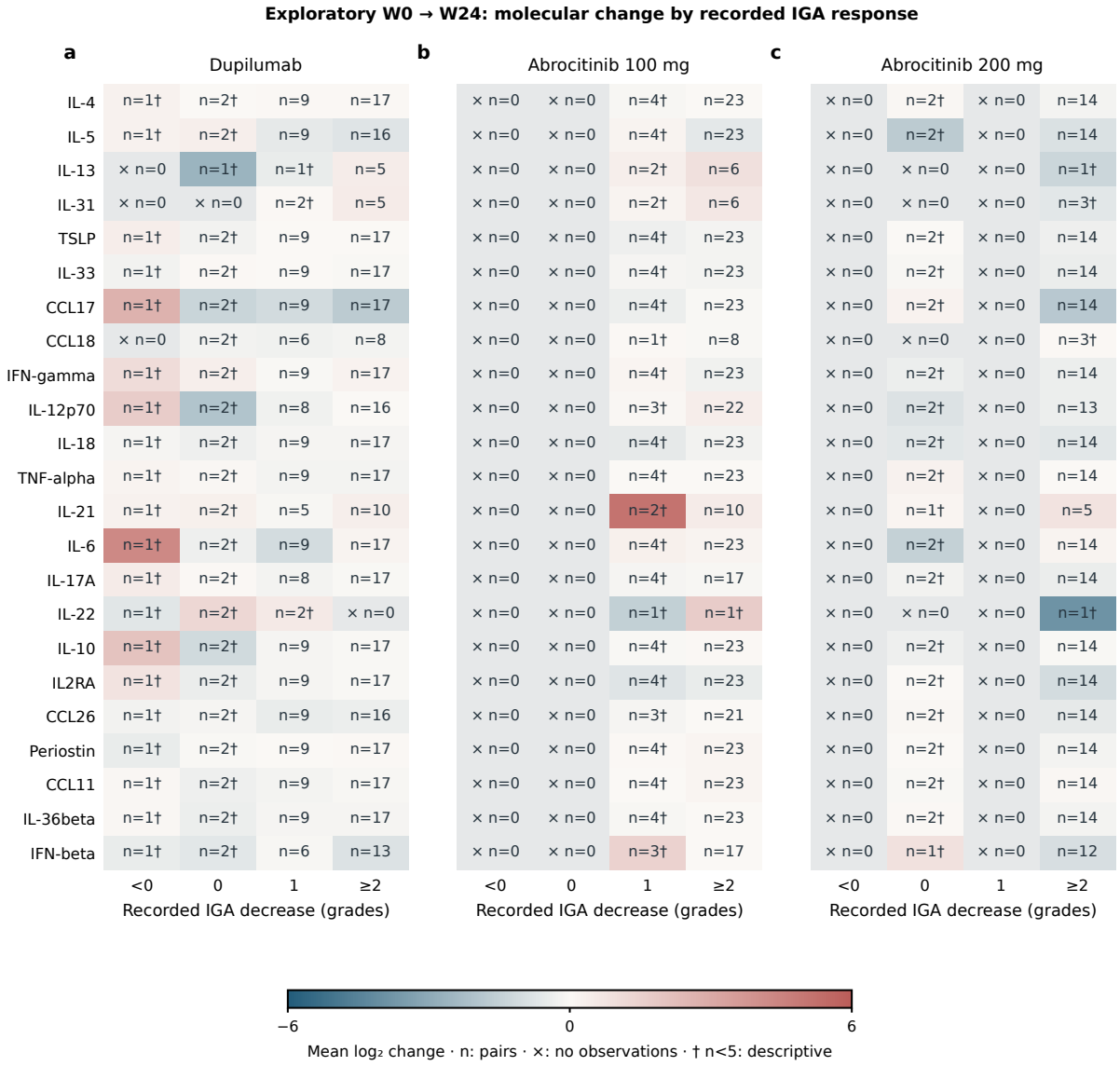


Figure S31: **Molecular changes by IGA response: W0 → W24.** Columns are mutually exclusive clinical-response categories within the named paired interval. Colour is mean within-patient $\log_2$ concentration change; numbers are complete-pair patient counts, with a dagger indicating descriptive small cells with n<5. Negative means lower later concentration. These are neither correlation coefficients nor change per clinical point. EASI improvement is relative to the start of the named interval; IGA uses recorded ordinal decrease. Cells marked × and n=0 have no observations; absence is not a zero molecular change. All 23 markers, exact means and intervals are retained in the source table; the small cells are descriptive and do not validate a prediction model.

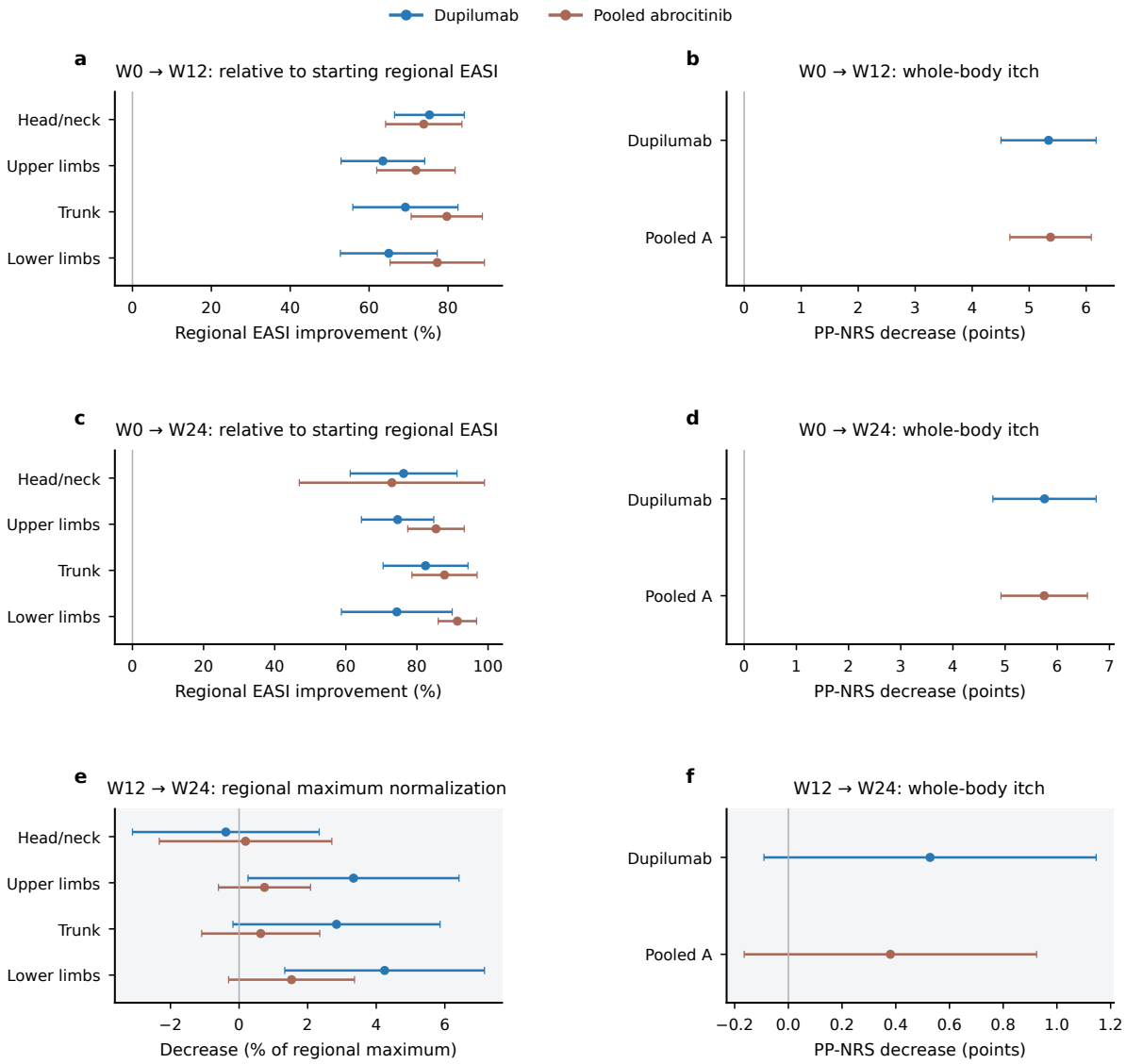


Figure S32: **Separate paired regional and whole-body itch intervals.** Rows are separate paired intervals that can share patients; they are not a continuous cumulative sequence. First two rows show mean regional percentage improvement relative to positive W0 regional EASI. Last row shows W12→W24 decrease as a percentage of the regional theoretical maximum, avoiding division by near-zero W12 values. Right, Whole-body PP-NRS decrease in original score points, not regional itch. Positive values indicate improvement, negative worsening. Bars are 95% t confidence intervals. Source-consistent regional records are required; the separate-dose and adjusted comparisons remain in the tables. No adjusted regional contrast passed the FDR threshold. The pooled abrocitinib group retains its observed dose mixture.

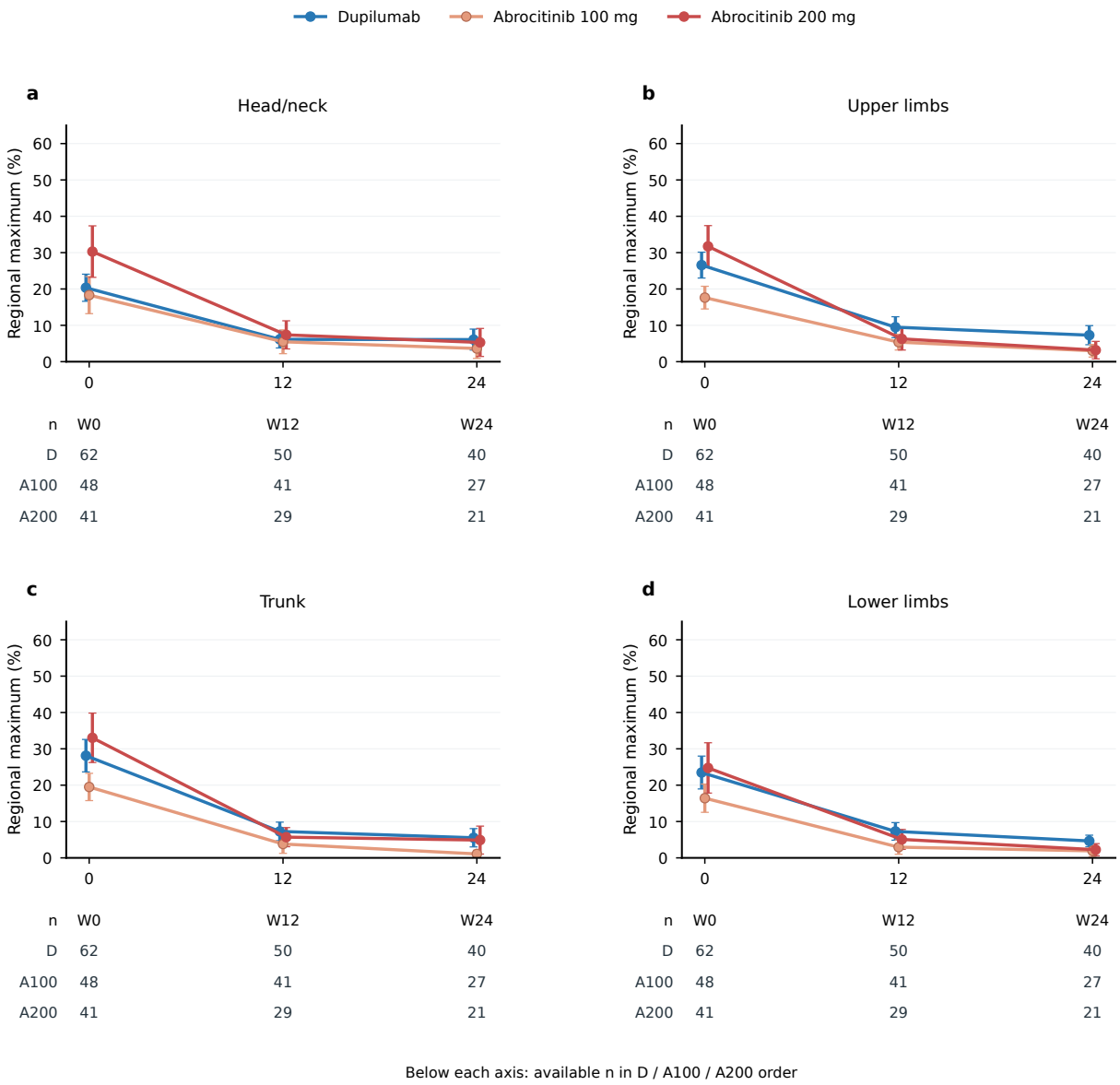


Figure S33: **Regional burden on a common interpretable scale.** Recorded regional EASI contributions divided by theoretical regional maxima (head/neck 7.2, upper limbs 14.4, trunk 21.6, lower limbs 28.8) and multiplied by 100. Points show observed means and 95% t confidence intervals, by initial treatment. This rescales skin-region burden; it does not measure molecular inflammation or demonstrate intrinsic anatomical susceptibility. Available patients may differ between visits. Exact n and sensitivity populations accompany the source table.

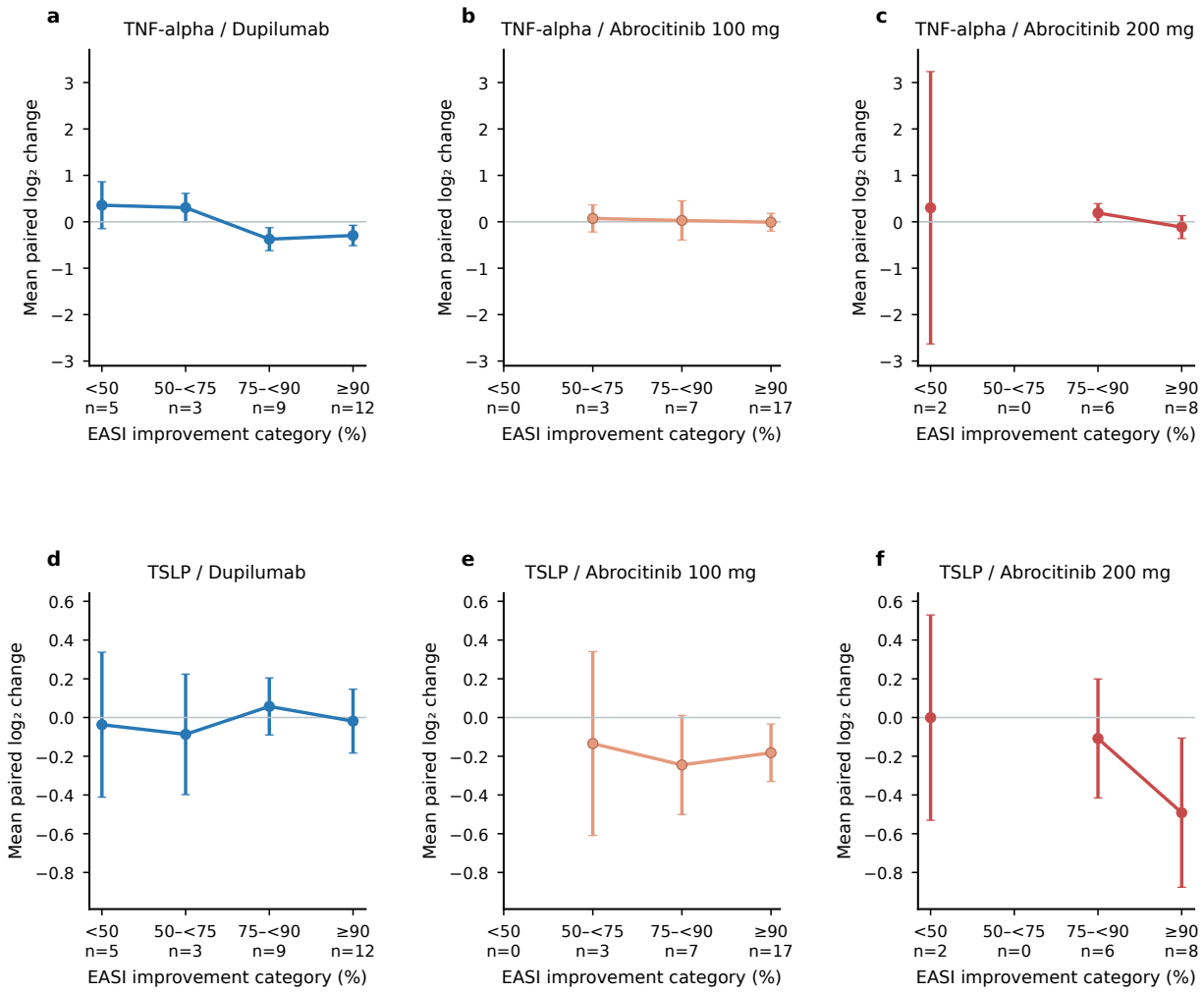


Figure S41: **EASI-associated molecular changes across response depth.** Exploratory descriptive plots for every marker whose change–EASI association met q<0.05 in the complete molecular–clinical test family (23 markers, five outcomes, three treatments and three windows). Both selected markers arise from W0–W24. Selection used the association family, not tests performed on these categories. All three initial treatments are shown. Points and available 95% t intervals summarize within-patient $\log_2$ concentration changes in mutually exclusive observed EASI-improvement categories; counts are printed below each category. Connecting lines join ordered categories, not visits. Sparse groups with fewer than five pairs are descriptive; absent intervals are not fabricated. The source tables retain all associations, including non-significant results. These same-data descriptions are not independent validation.

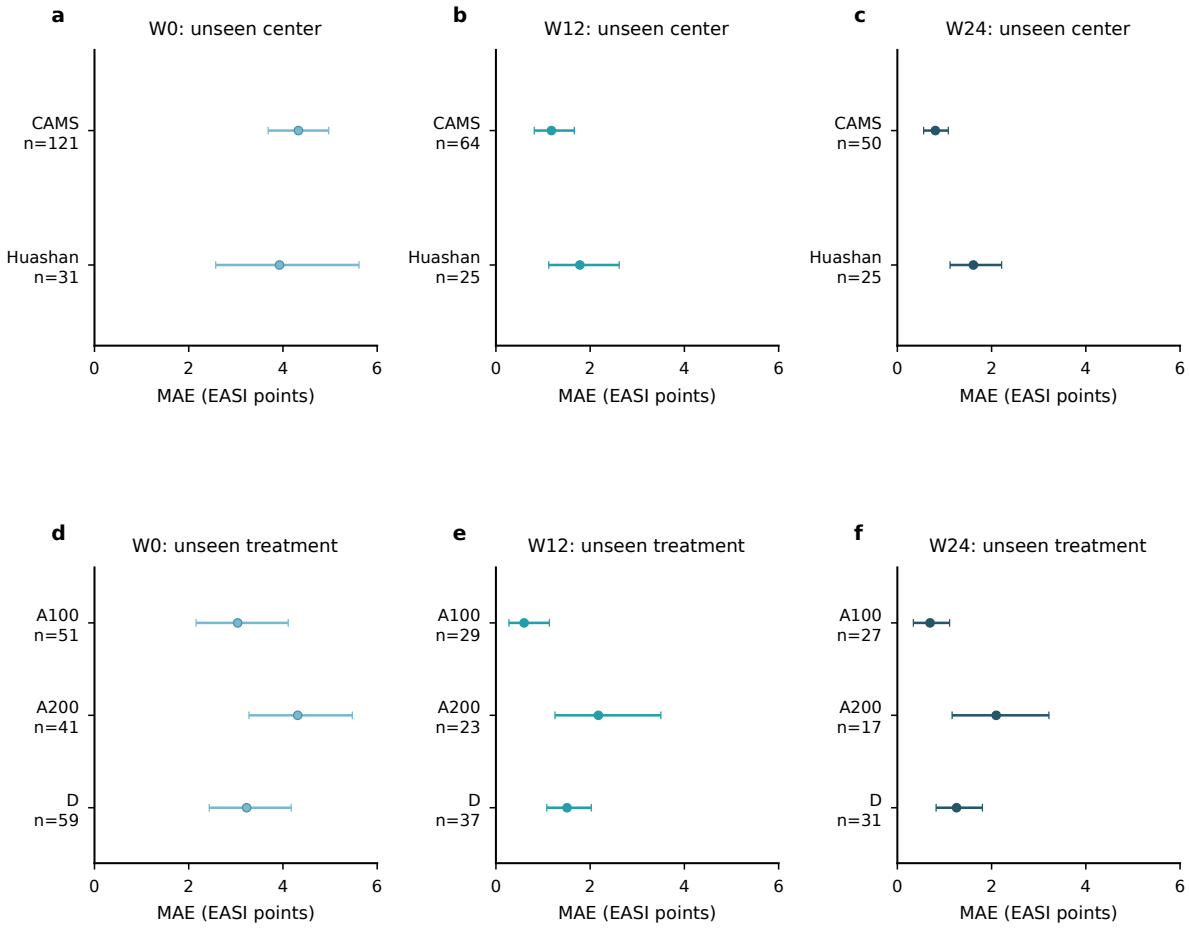


Figure S21: **Estimation when a centre or treatment group is left out.** Every displayed centre or initial-treatment group was entirely absent from fitting and model-form selection. Points are held-out MAE with patient-bootstrap 95% confidence intervals; patient counts appear next to each group. The same error scale is used in all panels. These internal transport stress tests are not independent external validation. Sparse groups and different severity distributions affect precision.

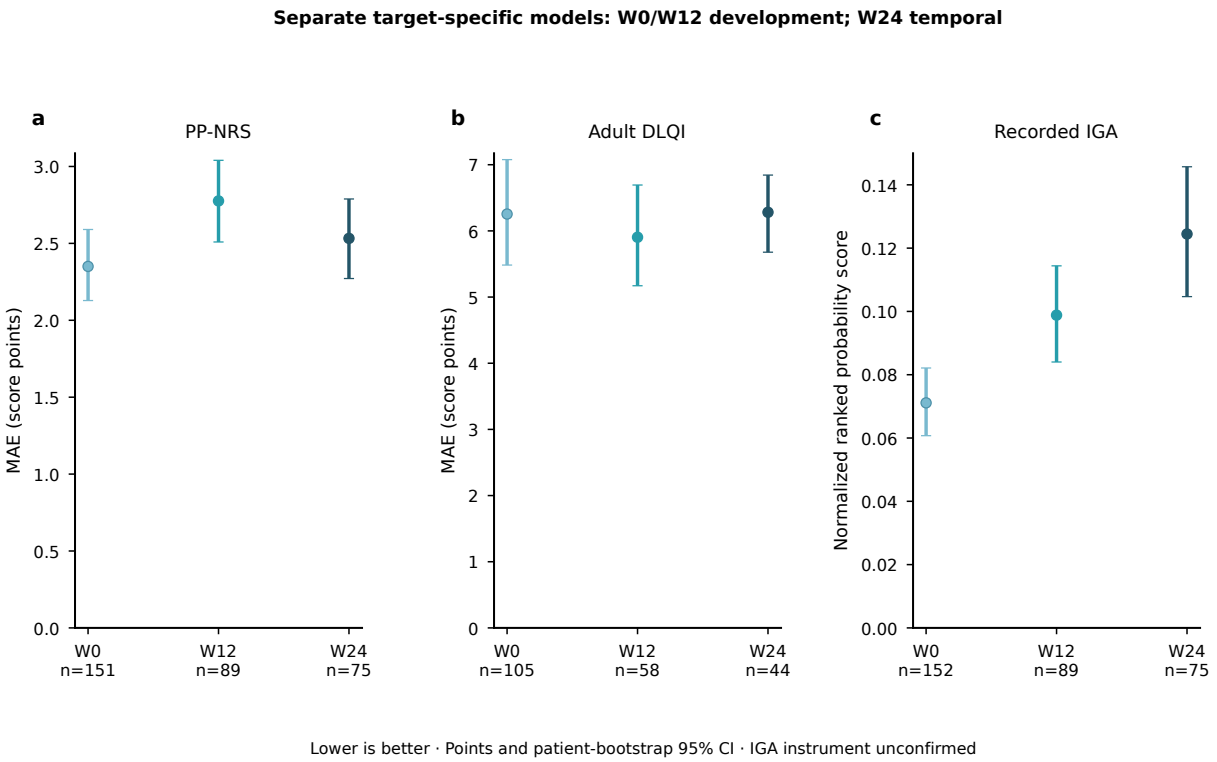


Figure S34: **Independently fitted PP-NRS, DLQI and ordinal IGA models.** Each target has separate coefficients and patient-grouped nested selection using current BSA and the fixed three-marker panel. PP-NRS and adult DLQI use continuous-outcome models; recorded IGA uses ordinal proportional-odds regression and normalized ranked probability score. Points and patient-bootstrap 95% confidence intervals show the independently fitted target-specific models; lower is better. W0, W12 and W24 use light blue, teal and deep blue circles. W0/W12 are development visits evaluated in patient-held-out outer folds; W24 is previously inspected same-cohort temporal transfer, not independent external validation. BSA-only controls remain in the numerical source table and are not plotted. No treatment, time label or EASI is used as an input. These results do not establish clinically adequate replacement, and the shared panel is not claimed optimal for every target. Recorded IGA includes grade 5 pending instrument verification.

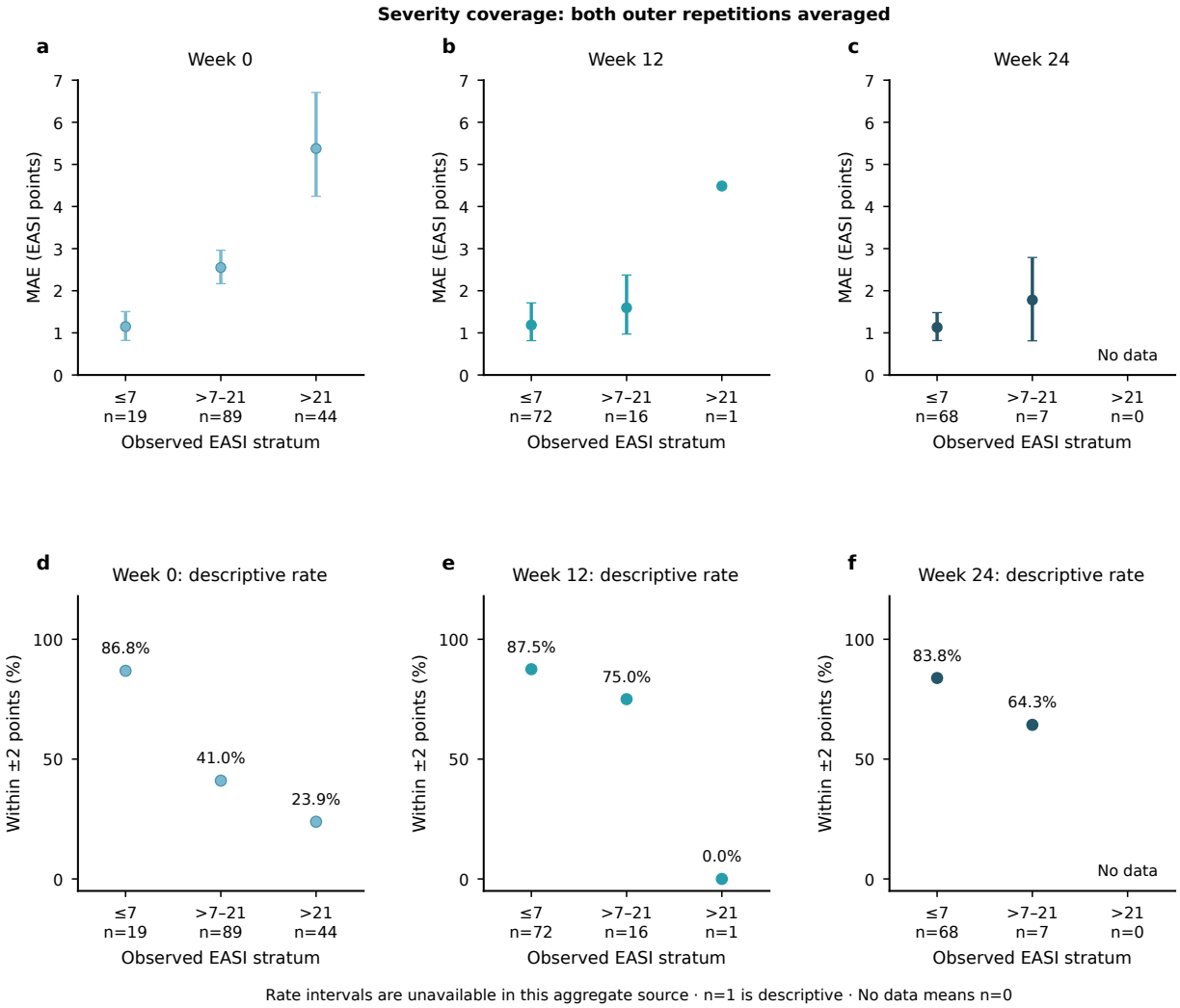


Figure S35: **Accuracy across observed EASI severity levels.** Primary nested estimator results averaged over both outer repetitions. Top, Mean absolute error and patient-bootstrap 95% confidence intervals. Bottom, Percentage within two EASI points, with direct values including zero; aggregate-source rate intervals are unavailable and are not fabricated. The n=1 result is descriptive. This is a descriptive accuracy threshold, not established clinical equivalence. Strata use observed EASI for evaluation only, not routing or predictor input. W12 has one severe patient and W24 none, so post-treatment accuracy cannot establish reliable severe-disease performance. Axes retain all interval extents.

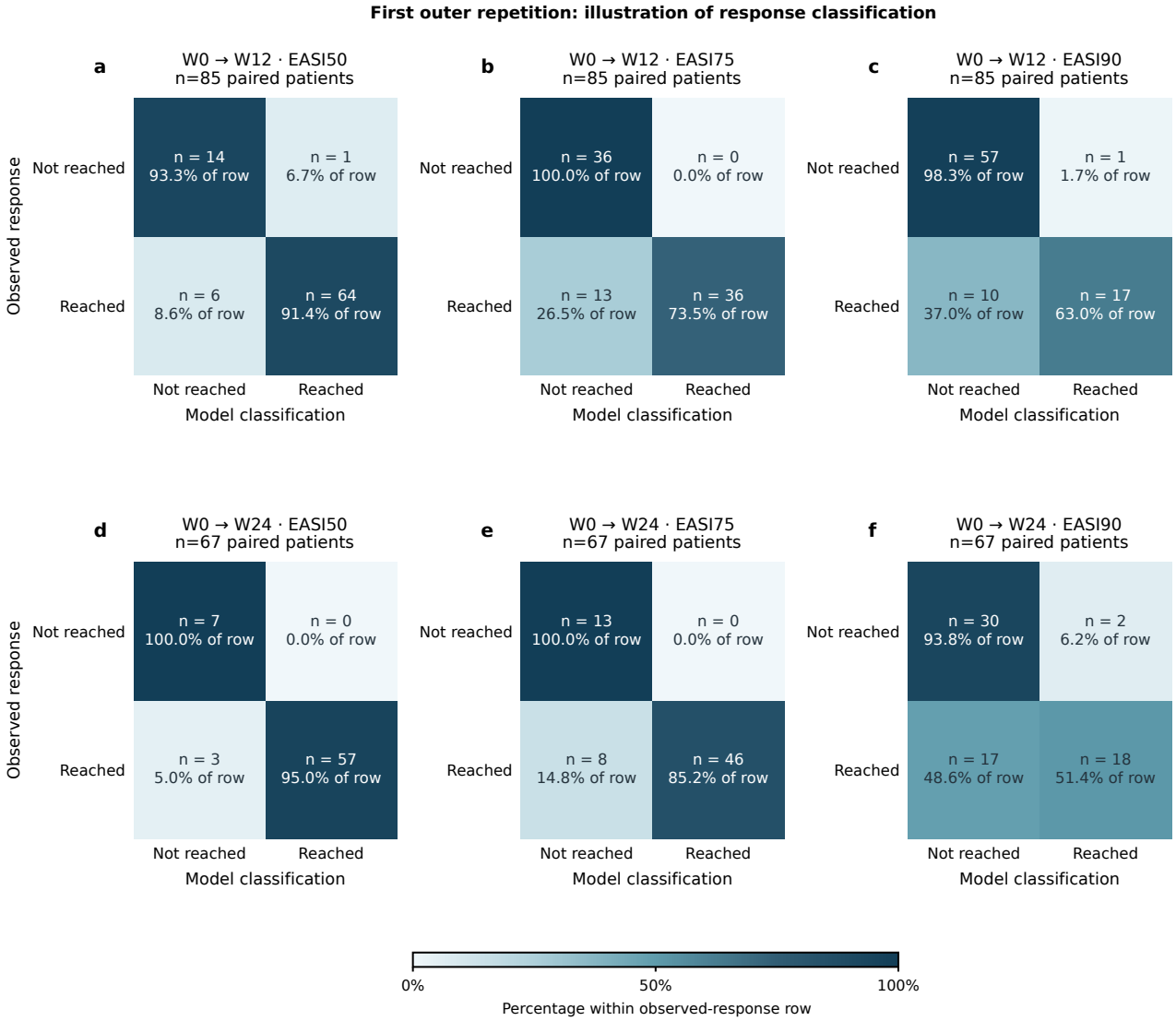


Figure S36: **Who is correctly classified as reaching EASI50, EASI75 and EASI90?.** Each panel shows actual patient counts from the first outer repetition, separately for W0→W12 and W0→W24. Rows are observed response and columns model-estimated response. Diagonal cells are correct; upper-right cells falsely indicate response and lower-left cells miss an observed responder. Percentages are within each observed-response row, so the lower-right percentage is sensitivity and the upper-left percentage specificity. Observed response uses observed baseline/follow-up scores, whereas estimated response uses predicted baseline/follow-up scores. Counts are integers from one repetition, not the primary two-repetition average; the full metrics table retains the latter. The paired denominators are 85 and 67, with no excluded pair for invalid baseline in this displayed repetition. The two intervals can share patients. This assesses response classification, not alternative-treatment outcomes or continuous-score equivalence.

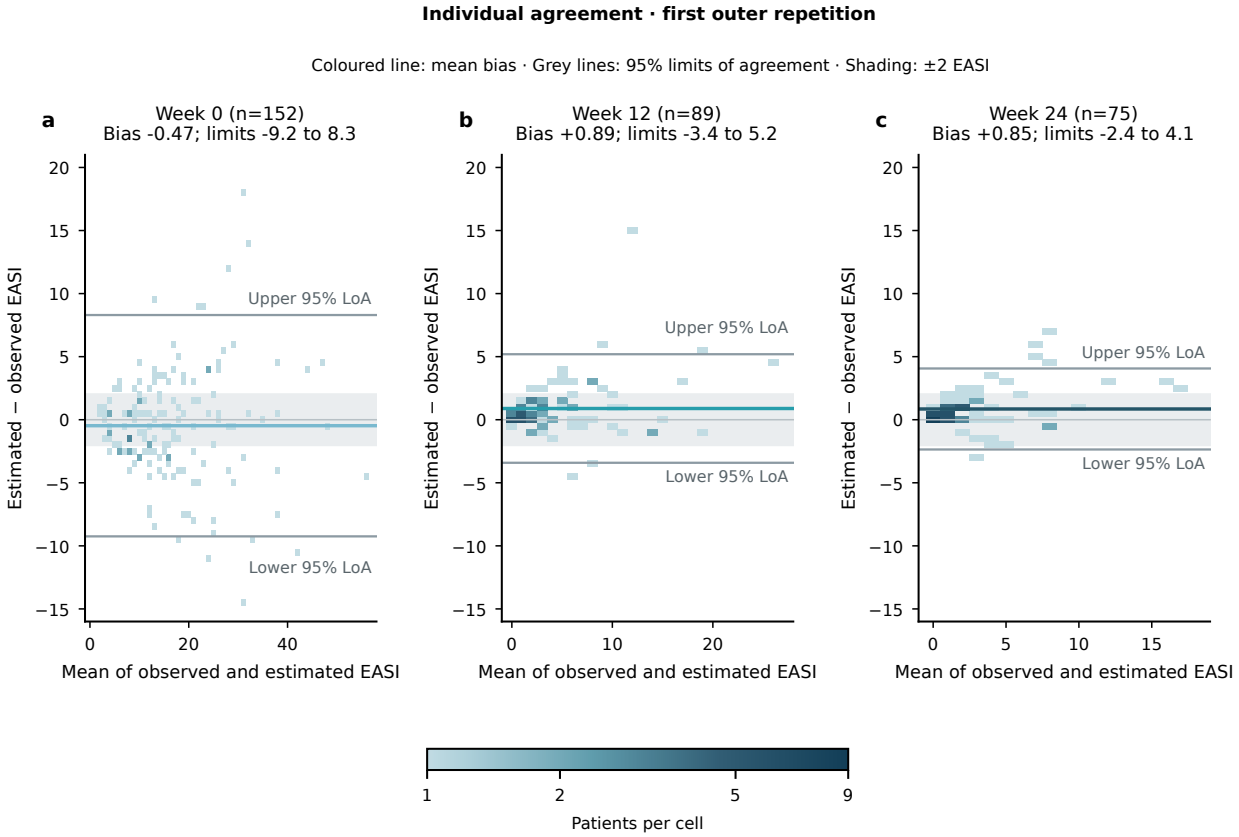


Figure S40: **Individual agreement across the range of EASI.** Bland–Altman count maps from the first outer repetition. Each equal-size cell spans one EASI point horizontally and half a point vertically; colour gives patient count on the labelled logarithmic scale. All 152/89/75 patients at W0/W12/W24 are retained. The vertical axis is estimated minus observed EASI; negative values indicate underestimation. Coloured horizontal lines show mean bias. Grey labelled lines are 95% limits of agreement (mean bias ±1.96 error SD), not confidence intervals. The common vertical scale contains every error and both limits. The ±2-point shaded band is a reading guide, not a validated equivalence margin. W0/W12 predictions are patient-held-out; W24 is same-cohort temporal application. Table 2 gives primary metrics averaged after scoring both repetitions.

Aggregate source tables

The machine-readable source-table index lists the complete aggregate collection, including unplotted and negative results. Figure-specific dependencies are in `figure_catalog.json`. The versioned aggregate directory includes clinical, molecular, model, regional and reference-sensitivity tables; restricted patient-level files are not part of the public collection.

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